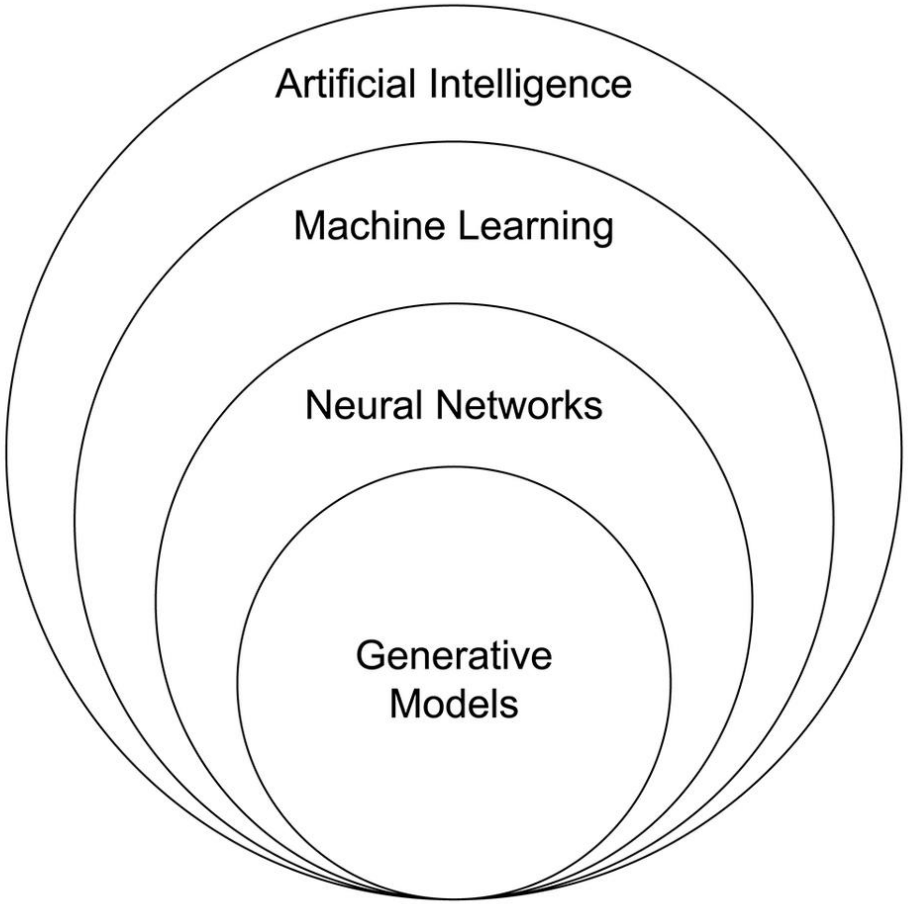
Machine Learning 1

Terminology

- AI
- ML
- Deep learning
- Statistical learning



https://commons.wikimedia.org/wiki/File:AI_relation_to_Generative_Models_subset,_venn_diagram.png

High-dimension datasets (n << p)

- SNP arrays
- RNA-seq
- Single-cell RNA-seq
- Methylation/epigenetics
- Proteomics
- Metabolomics
- Images
- Text/language
- …

2D plots are not enough

- Try exercises 1 and 2

The curse of dimensionality

Not just "there are a lot of variables, and we don't know which are important"

- Distance concentration
  - Points become ~equidistant
  - → "Nearest neighbor" means less
- Volume
  - Grows exponentially with p
  - → Data extremely sparse
  - → Huge sample size needed to cover the space
- Most of the volume is near the boundary
  - Very little interior
  - "Distance from center" means less
- Projections are deceptive
  - Lower-dimensional subsets will *appear* to have structure even if they're pure noise
- Statistical problems (more variables than datapoints)
  - → need for feature selection/extraction
  - → easy to find "perfect fit" based on noise alone (overfitting)

See Example 3

What kind of structure?

ML methods assume structure.

It’s up to you to ask the right question and choose the right tool.

- Are there clusters/groups?
- Does variation exist globally along smooth gradients?
- Is there predictive structure (i.e., can some variables predict others?)
- Are there local structures?
- Is there hierarchical structure?
- Are there networks/relational structures?
- How complex is the structure? What is the intrinsic dimensionality?

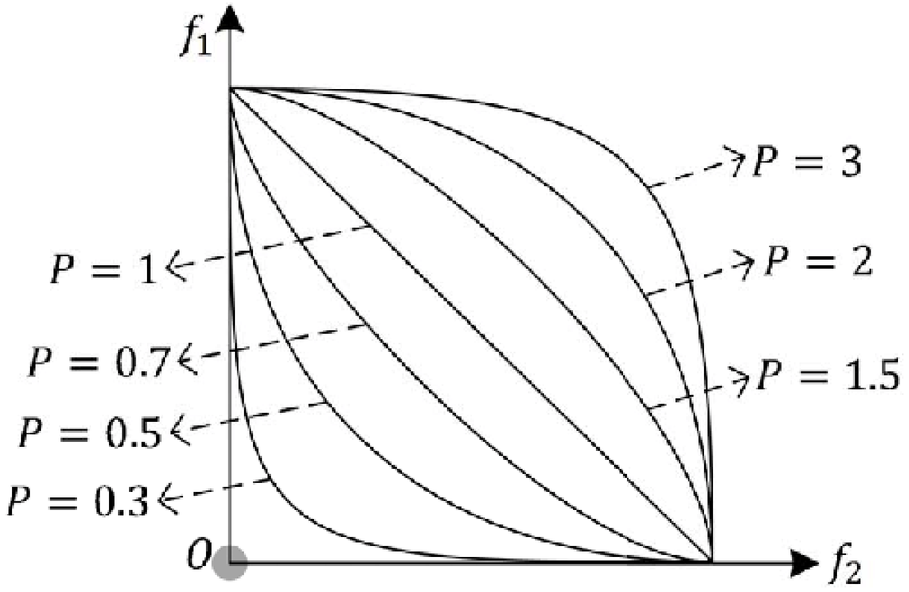
Supervised vs. unsupervised ML answer different questions

Type of ML	Kind of question	You are given	Goal	Assumptions
Supervised	Is there a mapping from X→Y?	• Input variables (X) • Target/outcome/labels (Y)	• Predict Y • Estimate relationships between X and Y	• The predictive structure (model type) • Signal in X could explain Y
Unsupervised	What structure exists in X?	• Only the data (X) • No outcome/labels	• Describe variation • Find groups, dimensions, gradients, etc.	• Continuous or discrete structure • Local or relational structure

How to measure distance

- Minkowski distance = $d_p(x,y) = \sum_{i=1}^{n} |x_i - y_i|^p$
  - Euclidean distance (p=2)
  - Manhattan Distance (p=1)
  - More generally, p >= 1
- Correlation distance
- Mahalanobis distance (analogous to a higher-dimensional z-score)

$$d_M(\vec{x}, \vec{y}; Q) = \sqrt{(\vec{x} - \vec{y})^\mathsf{T} \boldsymbol{\Sigma}^{-1} (\vec{x} - \vec{y})}.$$



Xu et al. 2018 IEEE Transactions on Cybernetics 49(11)

Euclidean distance example

Euclidean Distance b/w observation A and B in 2-D space (Feature 1 and 2)
= $\sqrt{(60-45)^2+(55-40)^2}$

Observation\features	Feature 1	Feature 2	Feature 3	Feature 4	Feature 5
Observation A	45	40	35	35	56
Observation B	60	55	50	50	71

Euclidean distance example

Euclidean Distance b/w observation A and B in 2-D space (Feature 1 and 2)

= $\sqrt{(60-45)^2+(55-40)^2}$

- Euclidean Distance b/w observation A and B in 5-D space (Feature 1~5)

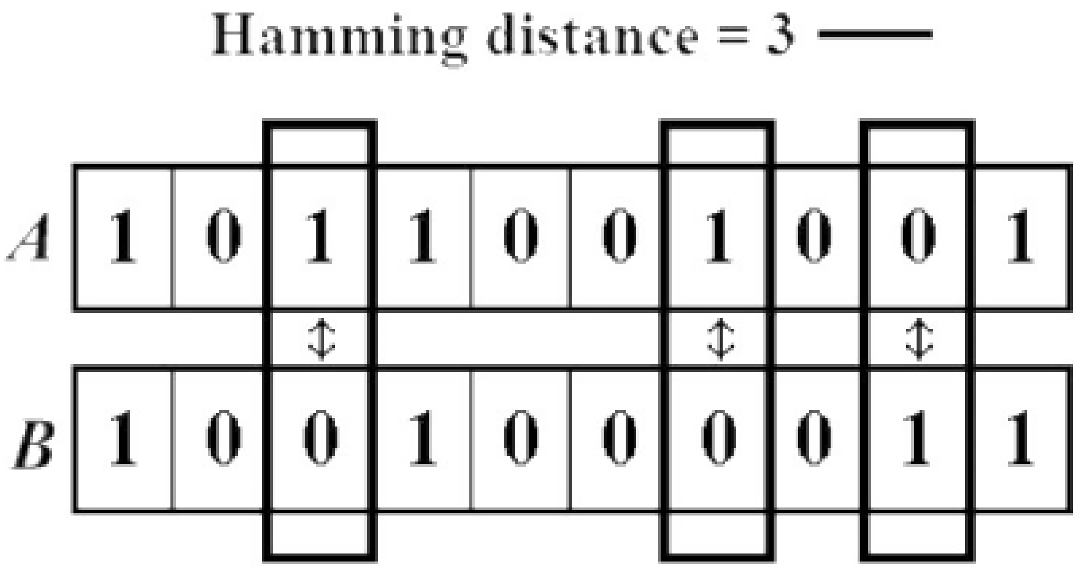
=

$$\sqrt{(60-45)^2+(55-40)^2+(50-35)^2+(50-35)^2+(71-56)^2}$$

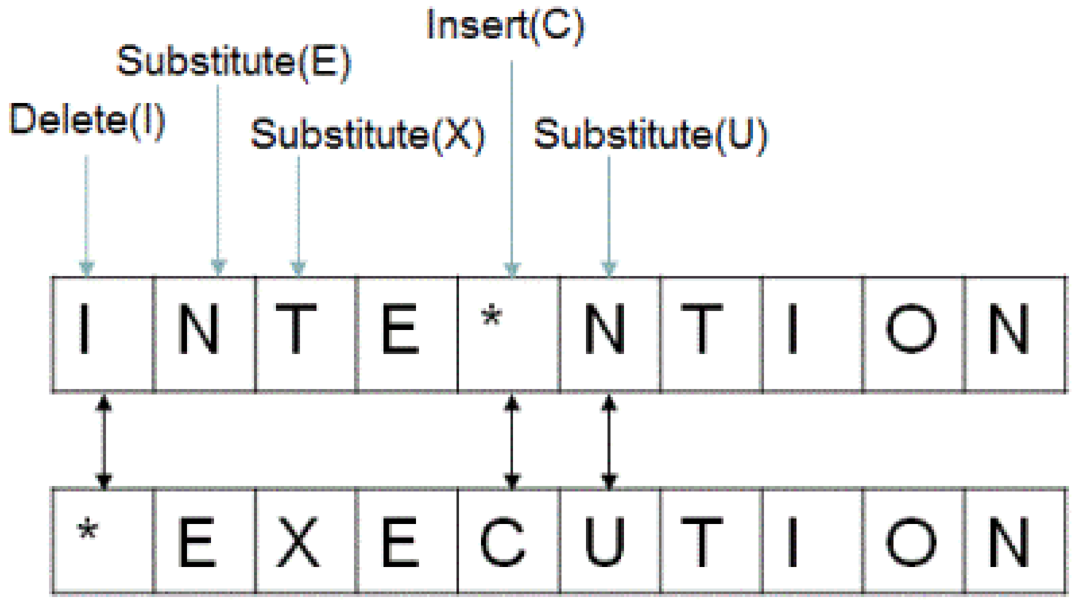
Observation\features	Feature 1	Feature 2	Feature 3	Feature 4	Feature 5
Observation A	45	40	35	35	56
Observation B	60	55	50	50	71

Other metrics

- Hamming distance
- Levenshtein distance



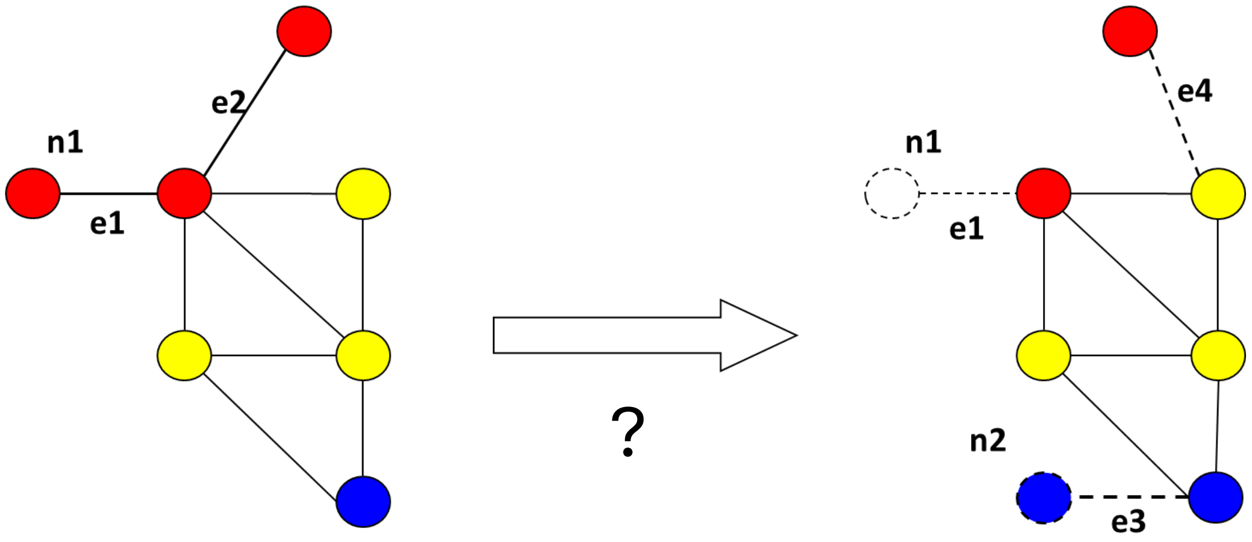
Ishengoma, DOI: 10.5815/ijieeb.2014.06.08



https://viblo.asia/p/edit-distance-L4x5xdQ15BM

Other distances (con’t.)

- The Edit Distance (Levenshtein distance) between two strings, str1 and str2 is the number of the number of operations required to transform str2 into str1
- Requires a set of operators
- E. g., graph edit distance with operators for
  - Edge add/delete
  - Node add/delete
- E. g., distance between two genotype sequences of the same length



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Unsupervised ML

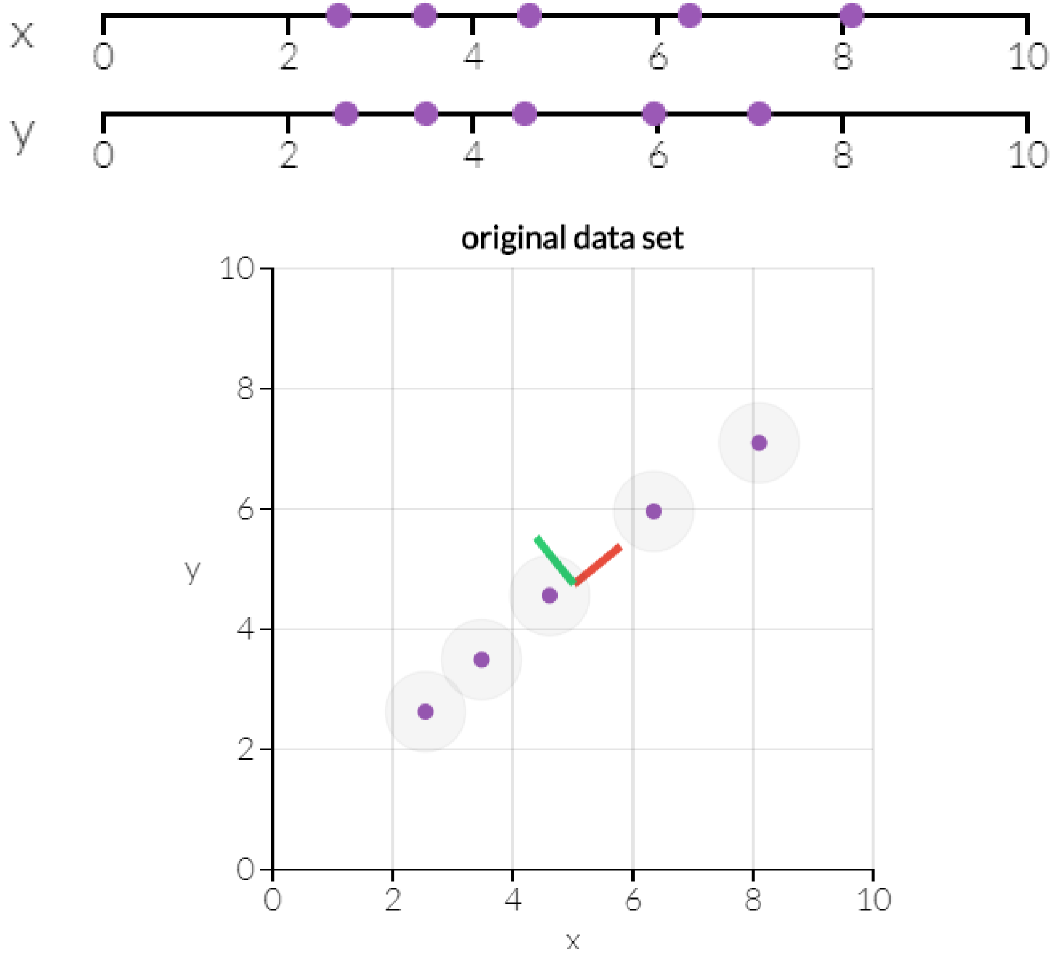
Main kinds of techniques, and some specific algorithms

- Dimension reduction
  - PCA, MDS, etc.
  - UMAP/t-SNE
- Clustering
  - Hierarchical clustering
  - k-means

Principal components analysis for dimension reduction

- How to find 2D (two variables) space in which information among data points are mostly kept?
- With variance in data as information, PCA can find a 1-d line preserving most information.

5 samples on a 2-d space (X and Y)

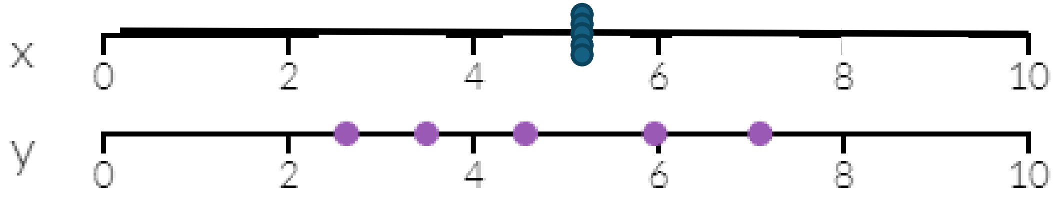


http://setosa.io/ev/principal-component-

Principal components maximize variance

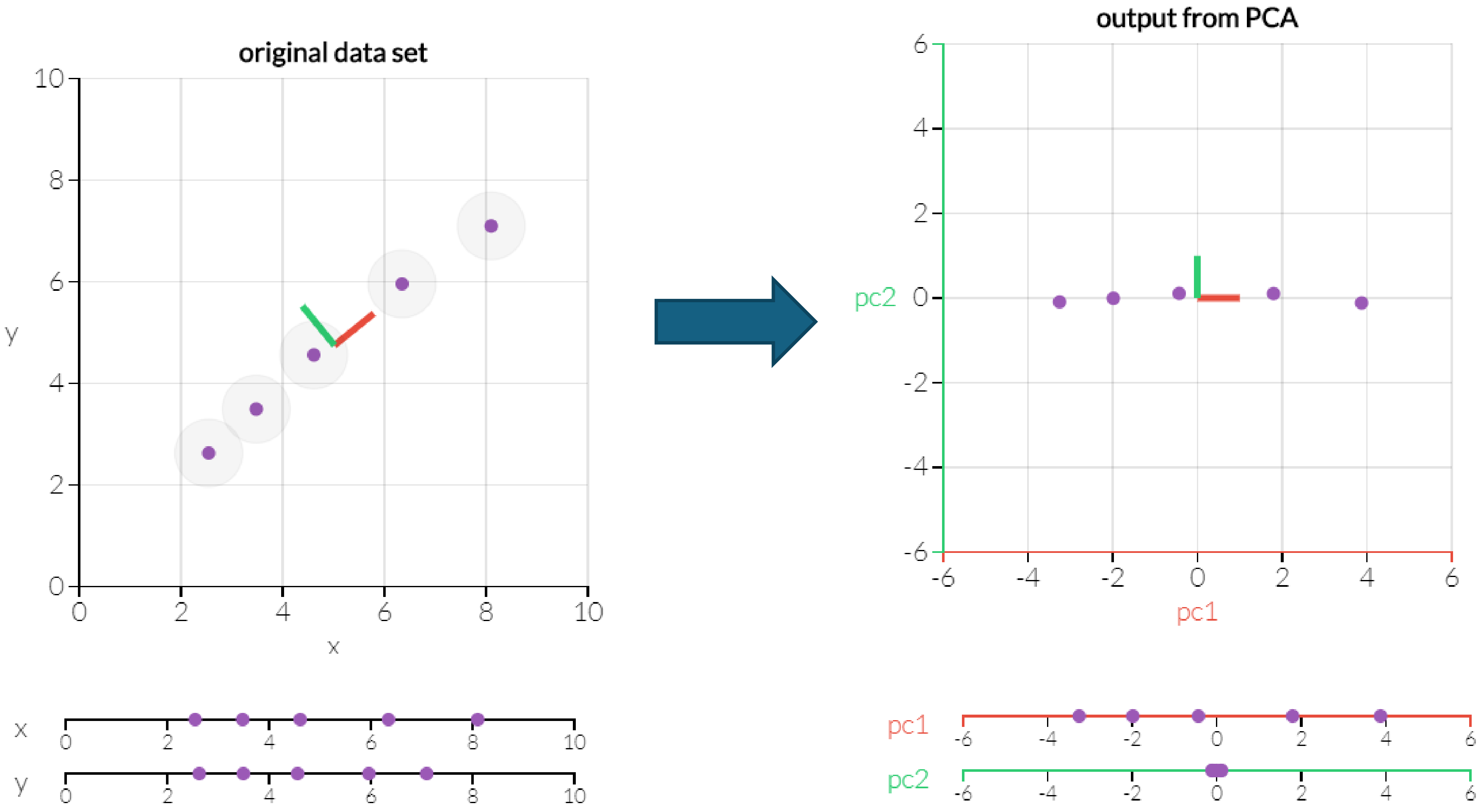
- How to find 2D (two variables) space in which information among data points are mostly kept?
- Variance in data can be taken as information

5 samples on a 2-d space (X and Y)



Adapted from http://setosa.io/ev/principal-component-analysis/

Principal components maximize variance

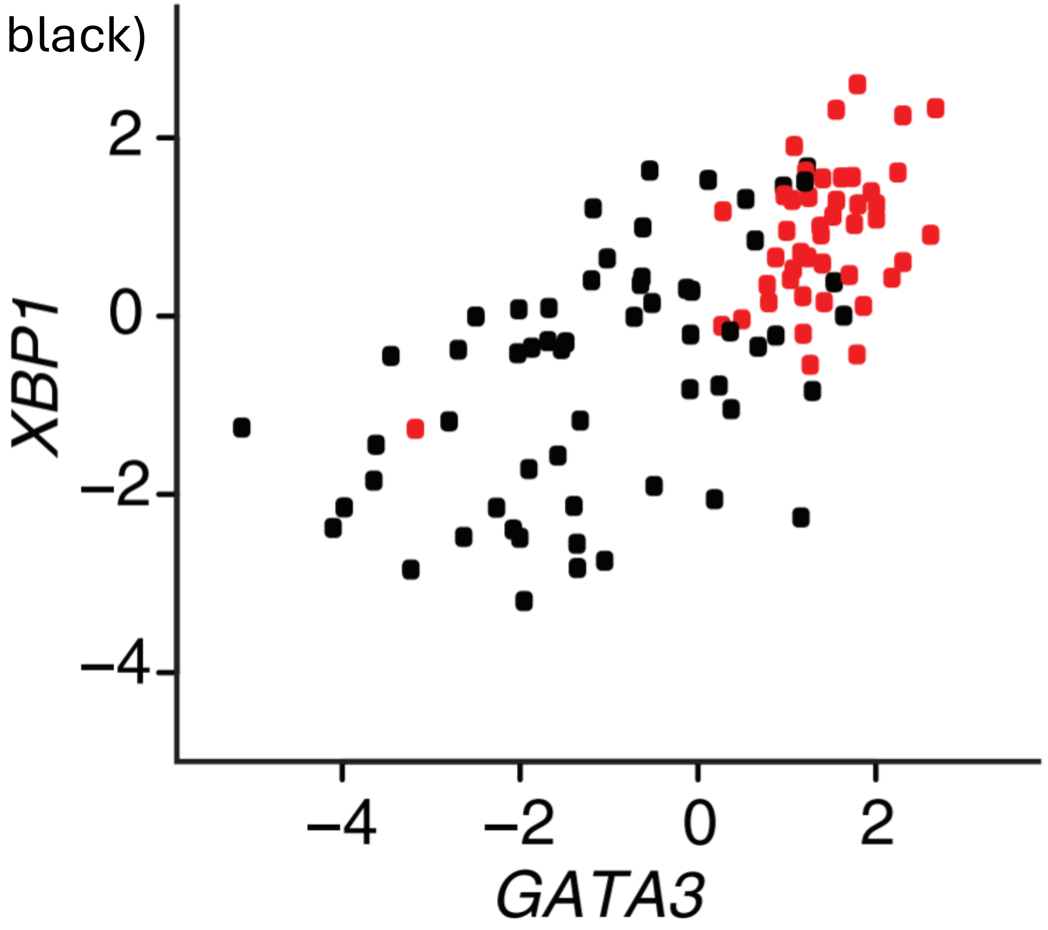


- Can identify two axes that maximize variance in high dimension

http://setosa.io/ev/principal-component-

Principal components in biomedical research

Breast cancer samples (ER+ in red and ER- in black)



Markus, Nat. Biotechnology, Vol. 26 Num. 3,

Principal components in biomedical research

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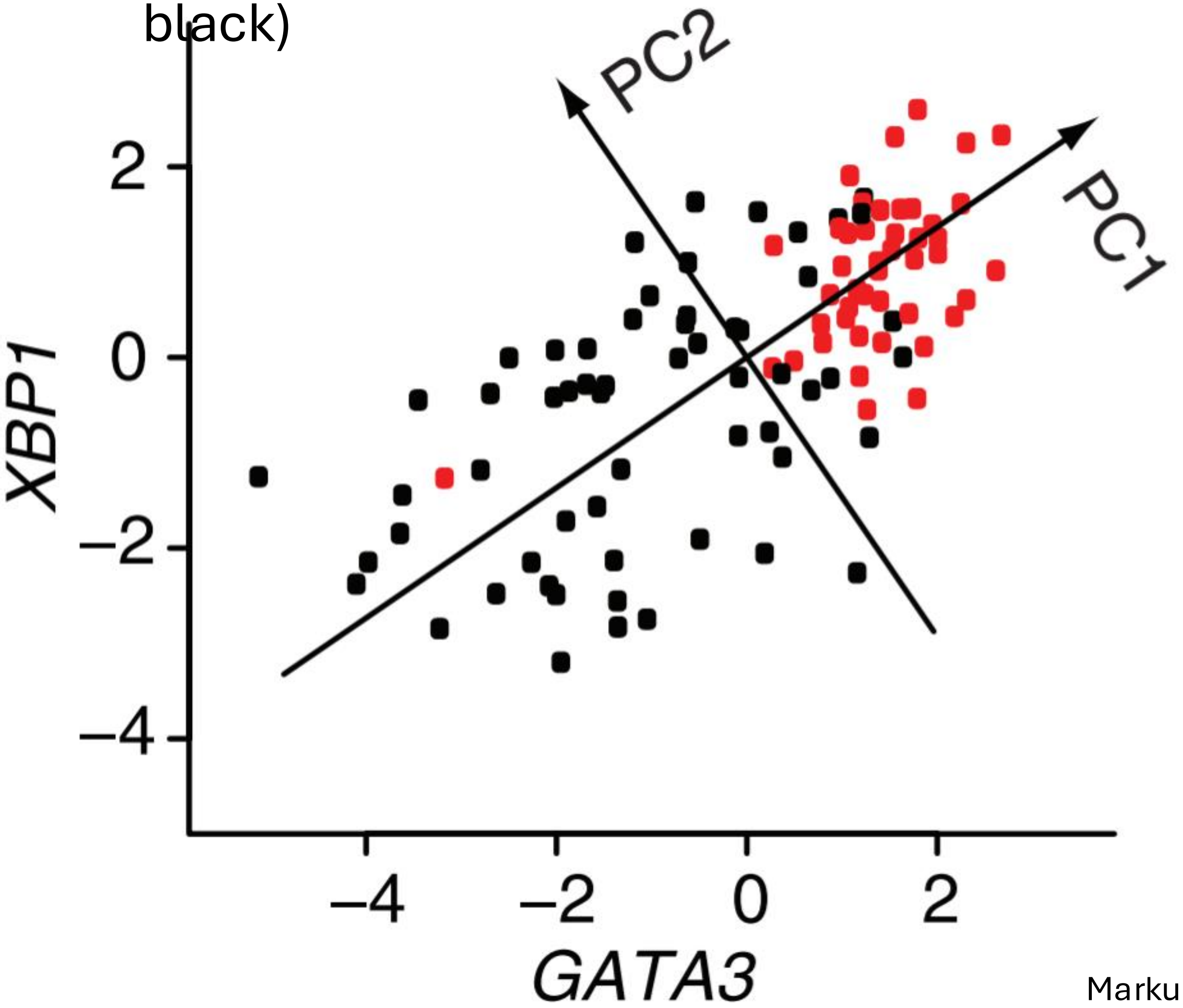
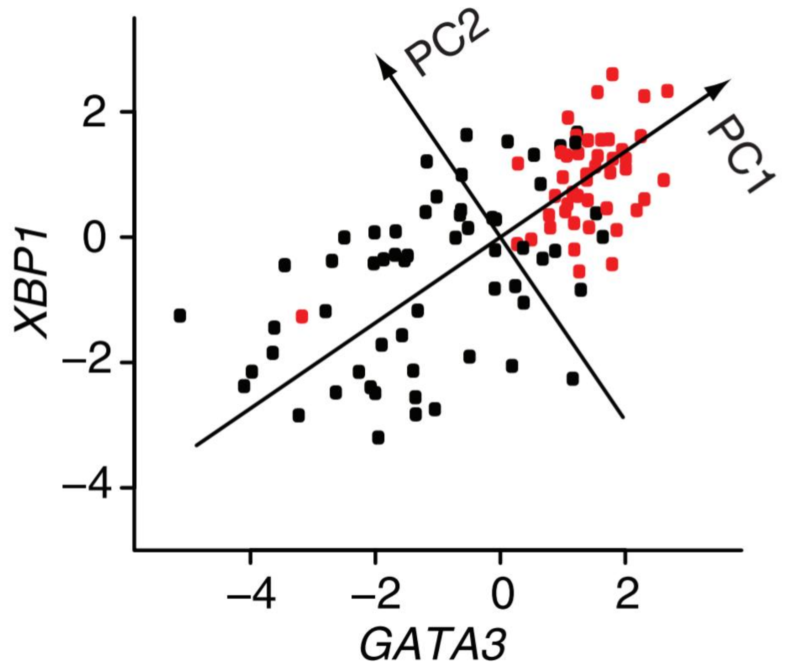


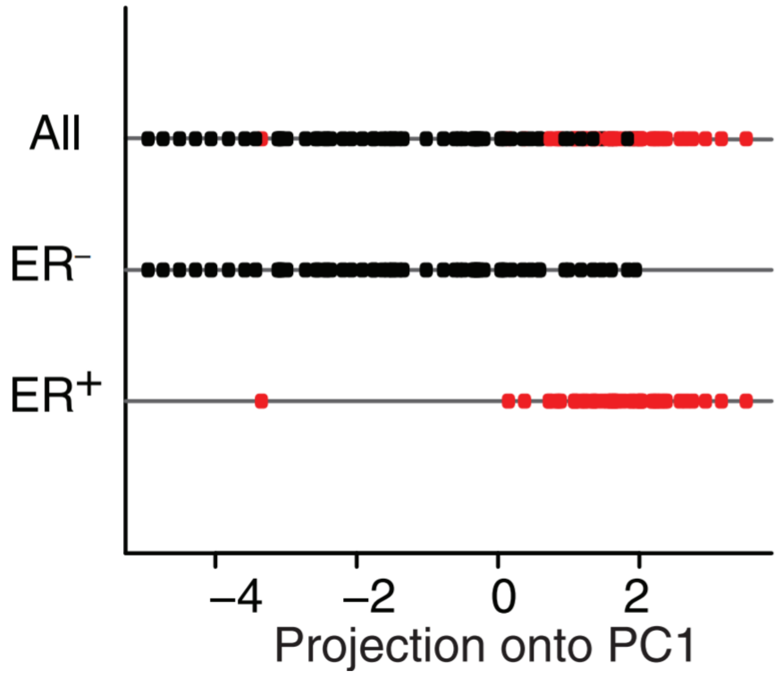
Markus, Nat. Biotechnology, Vol. 26 Num. 3,

Principal components in biomedical research

Things to analyze in PCA

1. Their projection to PCAs: PC1 well separates the grouping of your interest



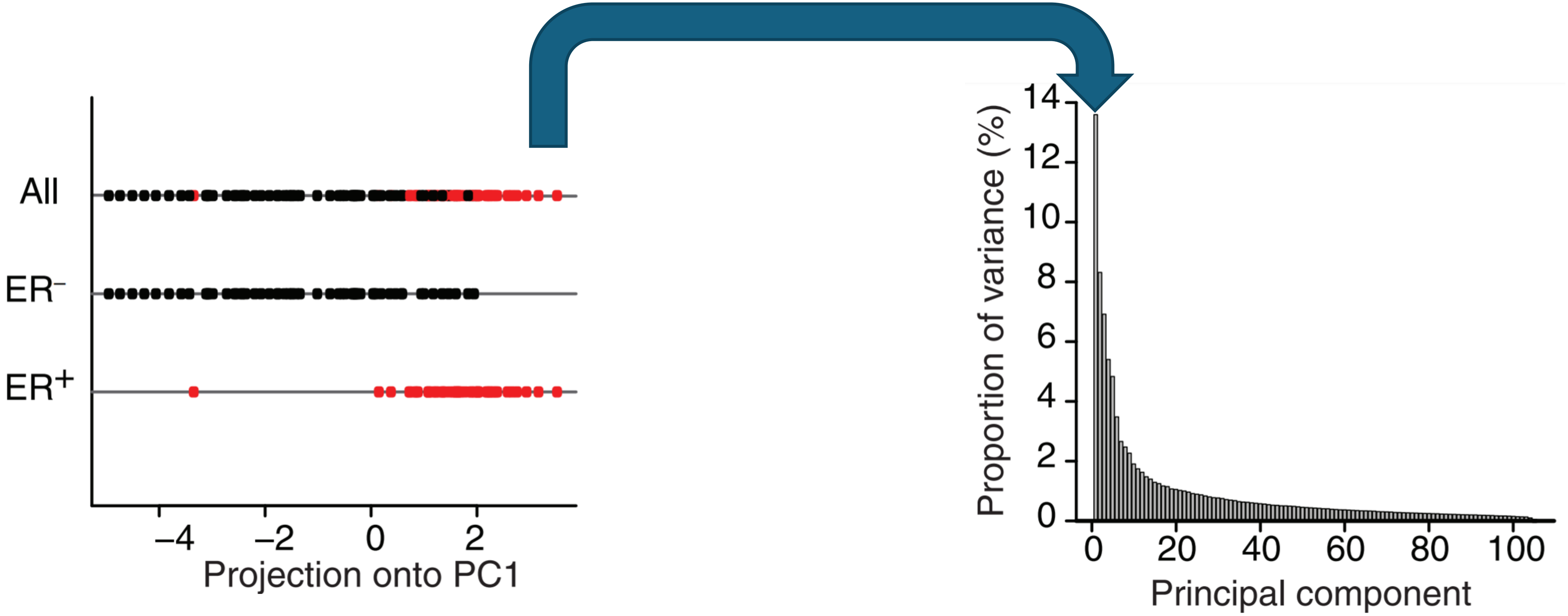


Markus, Nat. Biotechnology, Vol. 26 Num. 3,

Principal components in biomedical research

Some things to check in PCA

1. Projection of data onto PCs
2. Proportion of variance per PC

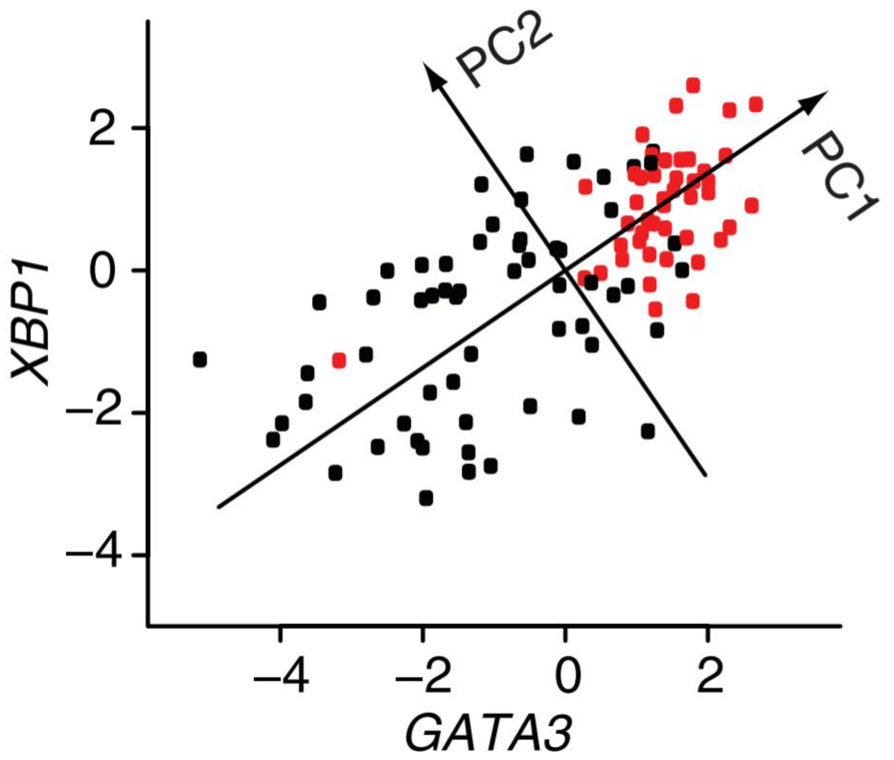


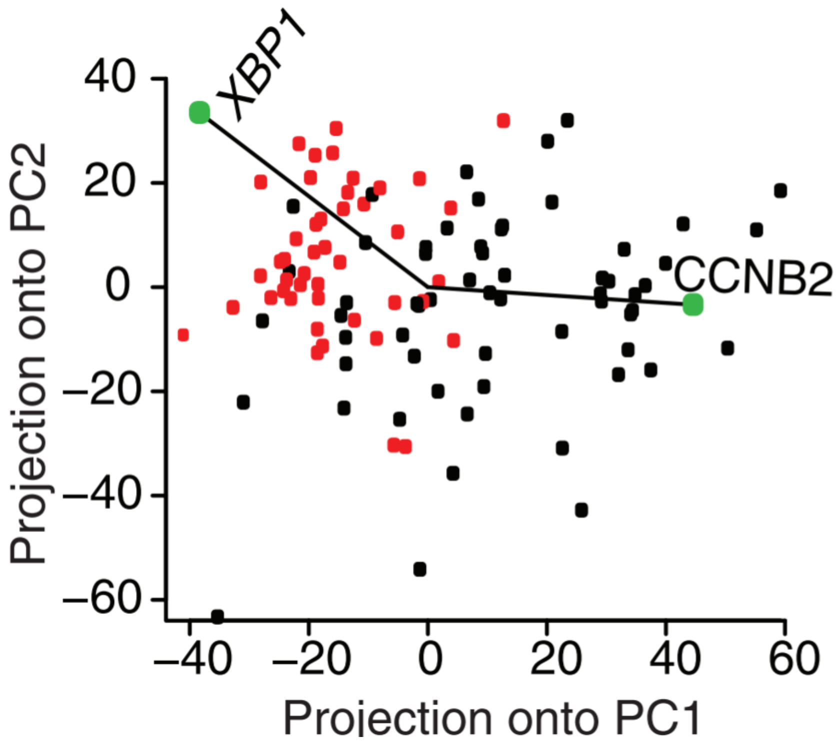
Markus, Nat. Biotechnology, Vol. 26 Num. 3,

Principal components in biomedical research

Things to check in PCA

1. Projection of data onto PCs
2. Proportion of variance per PC
3. Weights of your genes on the PCs



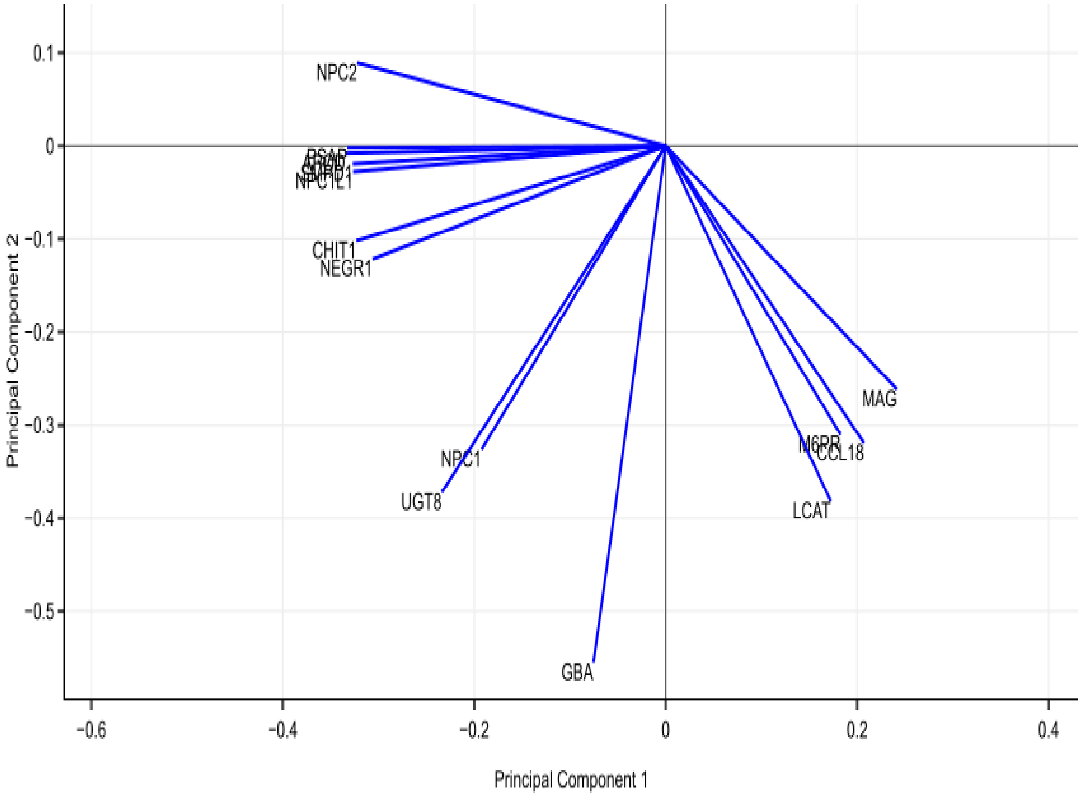


Markus, Nat. Biotechnology, Vol. 26 Num. 3,

A loading plot shows how strongly each dimension influences a principal component.

A loading plot shows

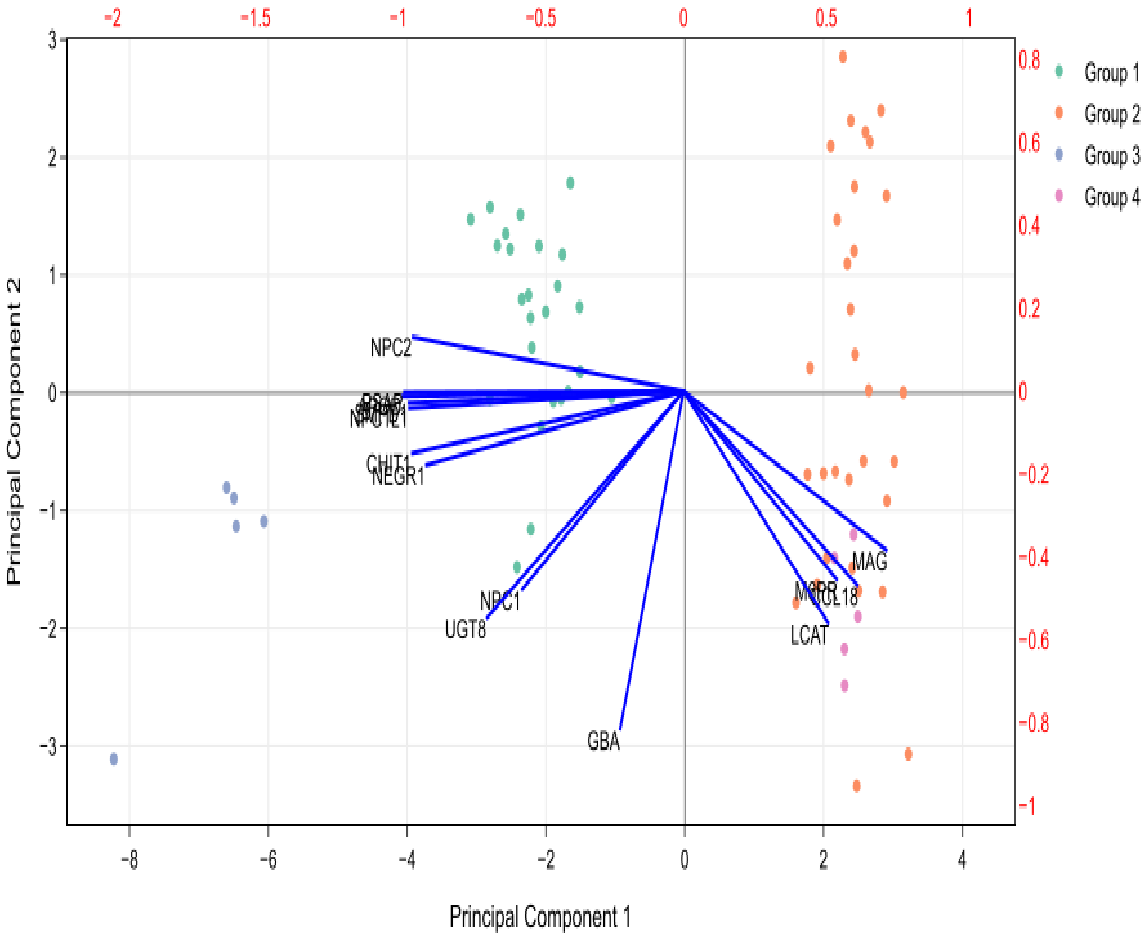
- Their projected values on each PC show how much weight they have on that PC
  - e. g., NPC2 and MAG influence PC1
  - e. g., GBA influences PC2
- The angles b/w variables show how they correlated with each other
  - e. g., APOD and PSAP positively correlated
  - NPC2 and GBA not likely to be correlated
  - NPC2 and MAG negatively correlated



http://setosa.io/ev/principal-component-analysis/

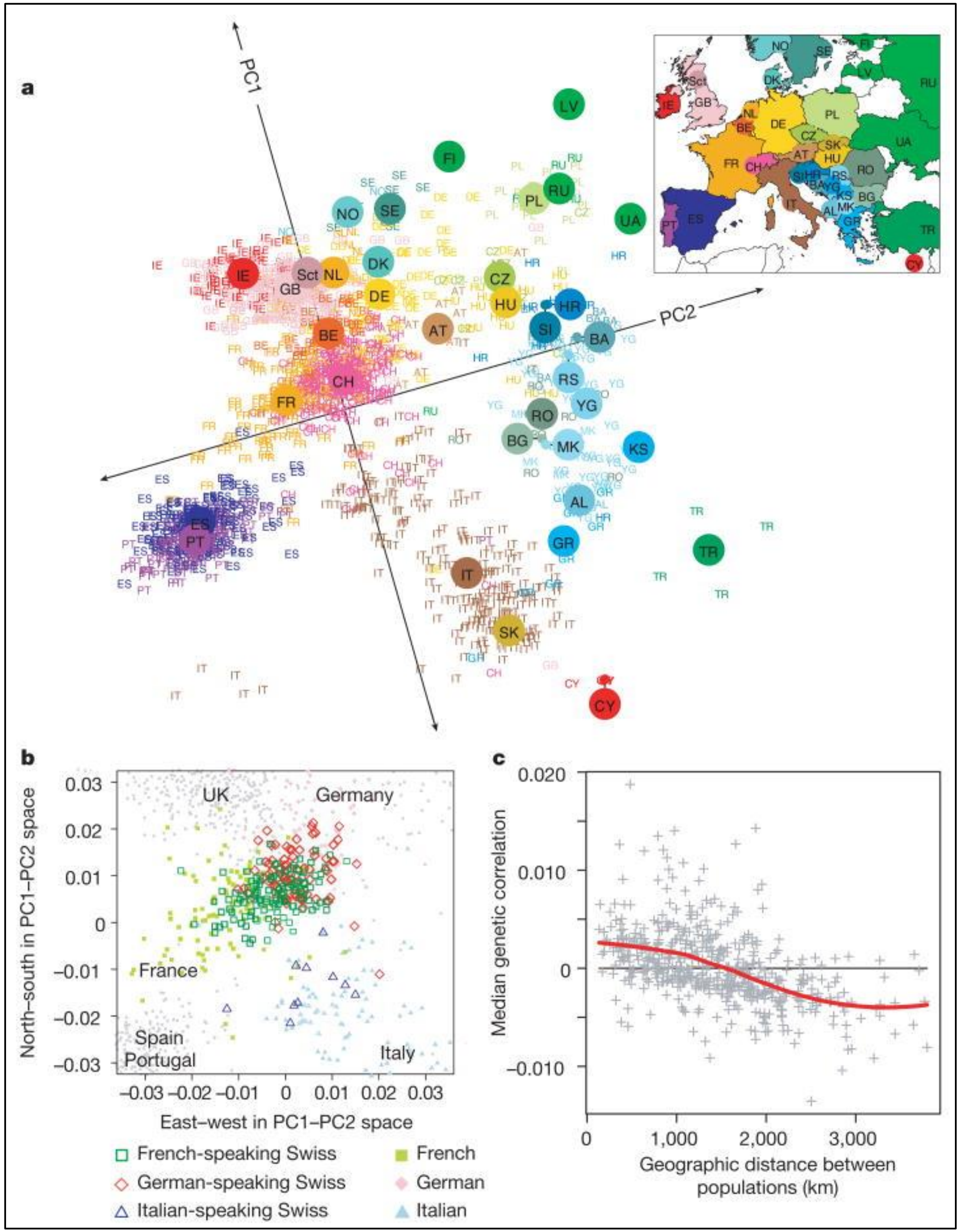
PCA biplot = PCA score plot + loading plot

- Bottom axis: PC1 score
- Left axis: PC2 score
- Top axis: loadings on PC1
- Right axis: loadings on PC2



http://setosa.io/ev/principal-component-analysis/

Example of PCA in genetics: principal components of ancestry



Limitations of PCA

- PCs are <u>linear</u> combinations of data
- Outliers
- Missing data

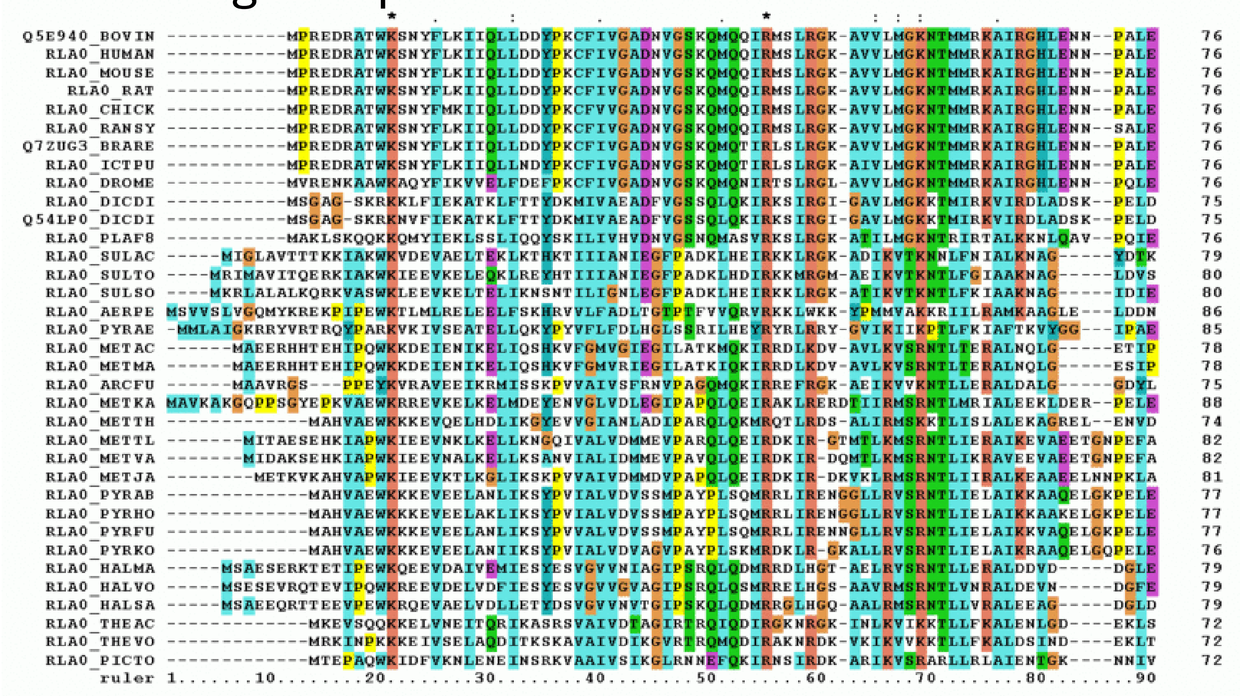
PCA examples

See exercises 4-8

Hierarchical clustering groups data points in a nested fashion by similarity

1. What data was used?

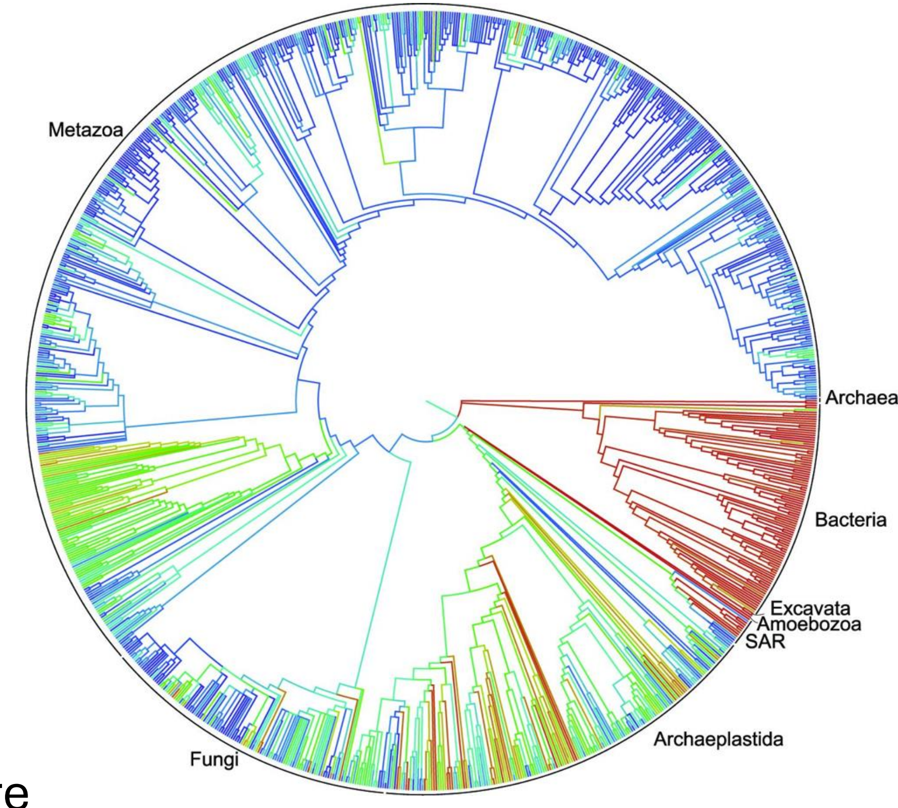
DNA sequences of biological species



https://en.wikipedia.org/wiki/Multiple_sequence_alignment

2. How was the distance estimated? A distance measure

Clustering of biological species



Hierarchical clustering groups data points in a nested fashion by similarity

1. What data was used?

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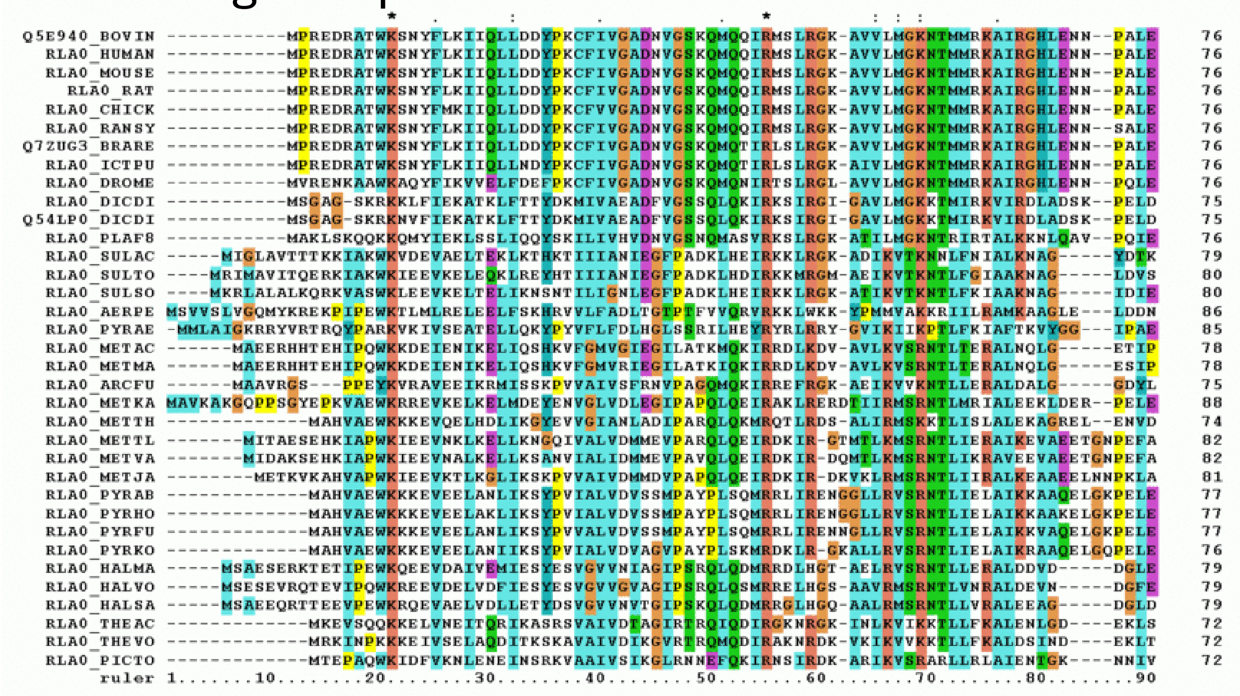
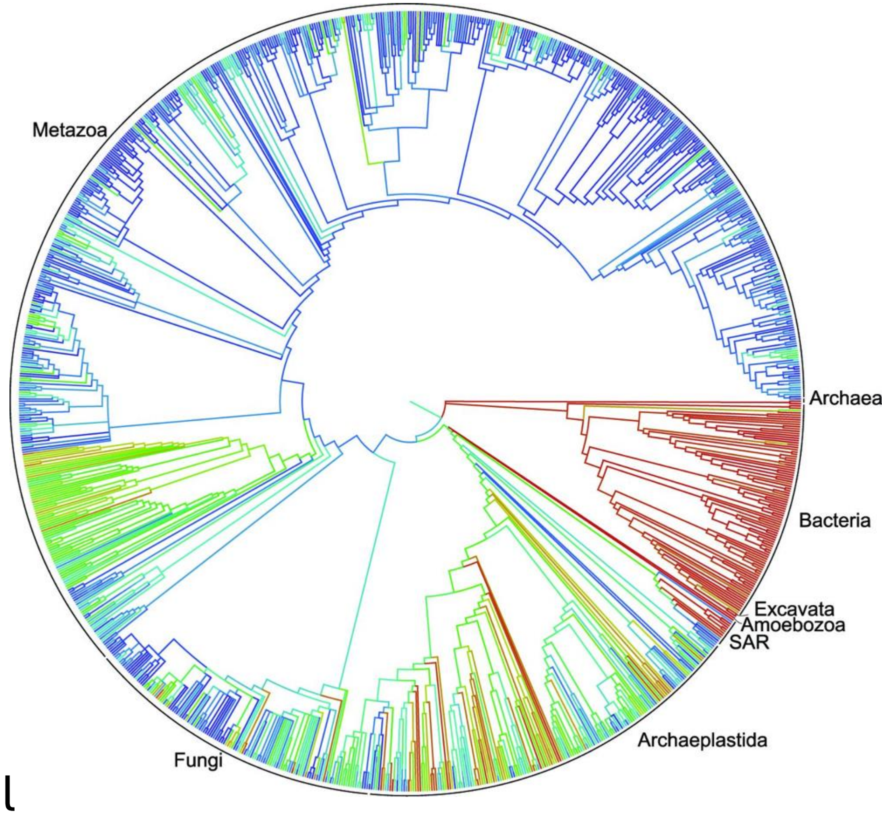


https://en.wikipedia.org/wiki/Multiple_sequence_alignment

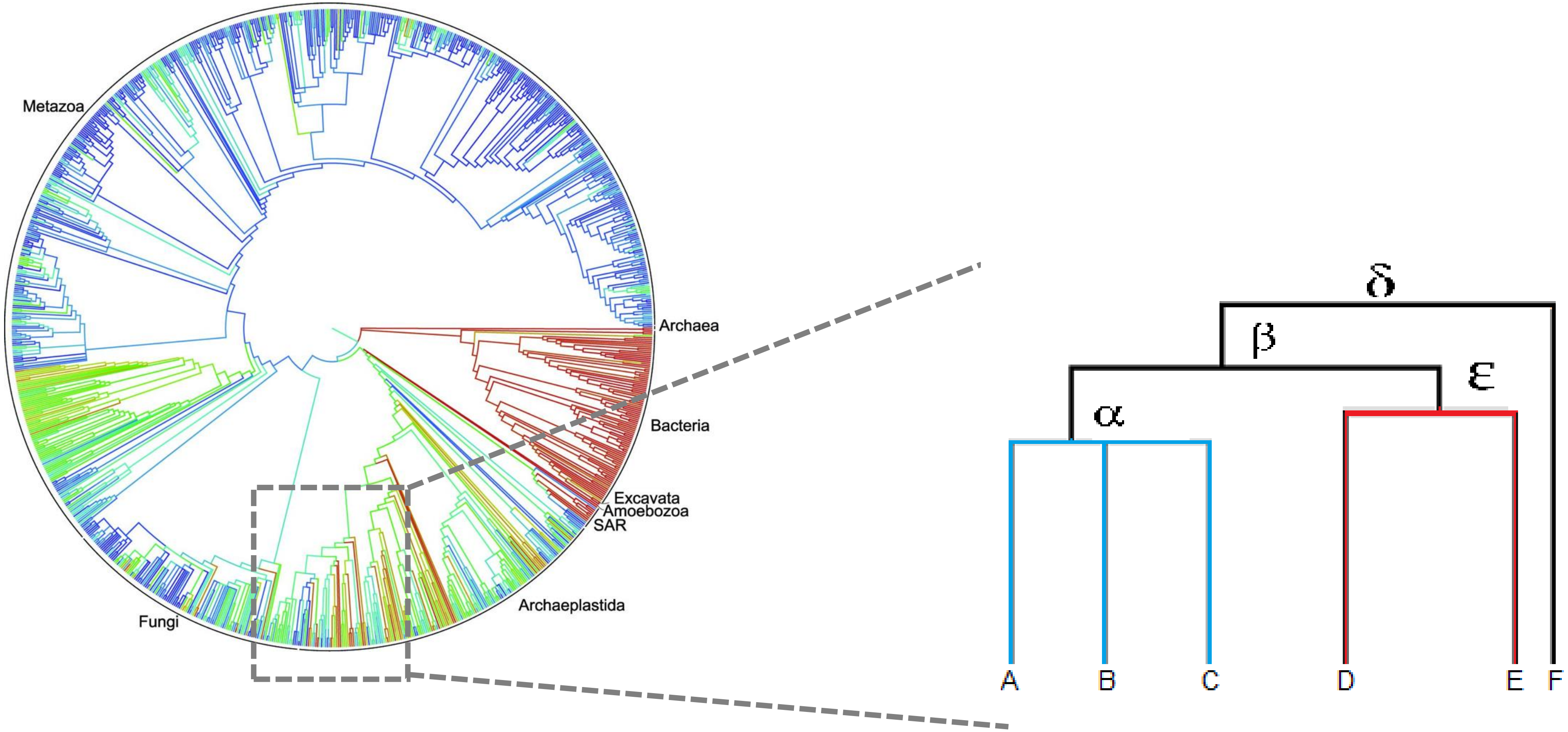
2. How was the distance estimated? An evolutional model

A phylogenetic tree



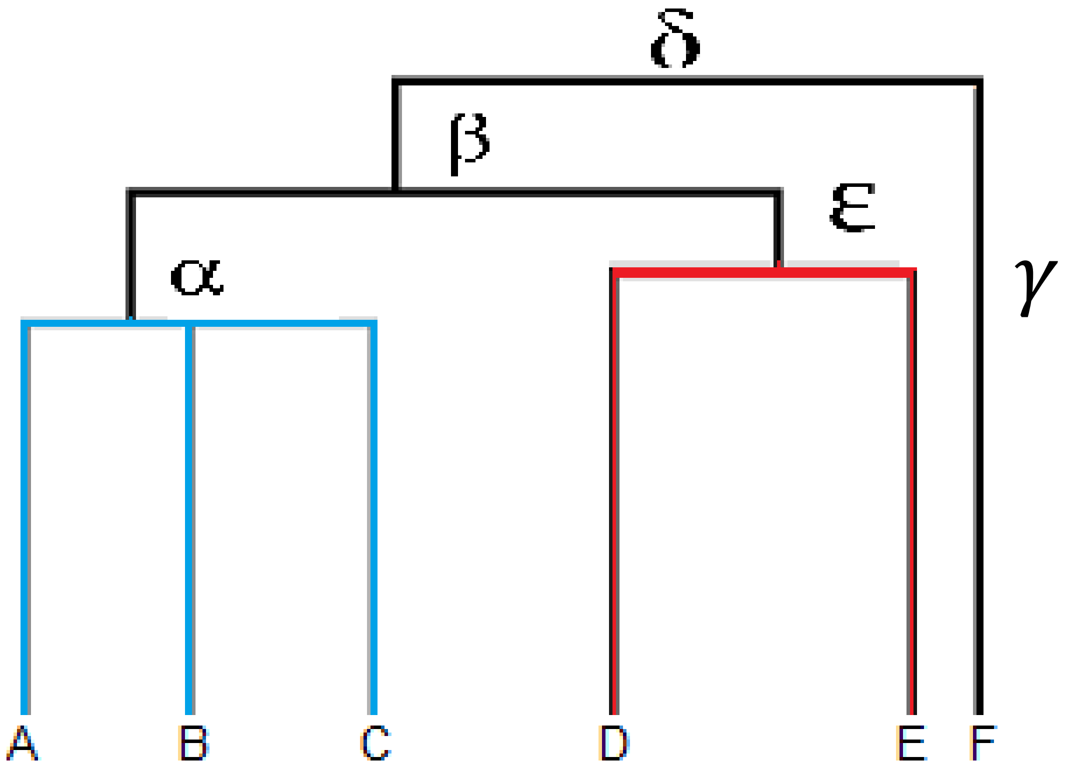
Hierarchical clustering makes a tree (dendrogram)

Clustering of biological species



Dendrogram visualizes hierarchical relationship of terminal nodes

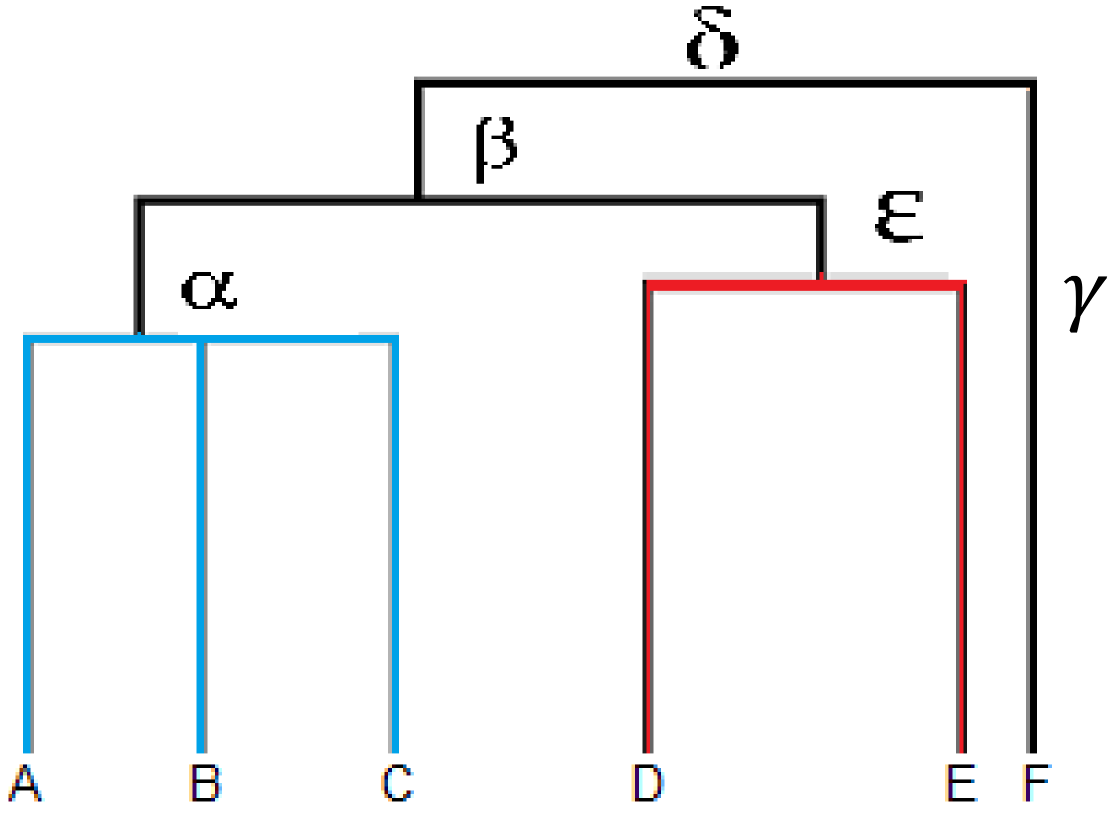
- Dendrogram = nodes + edges
- Nodes
  - root node: no parent allowed
  - intermediate nodes: multiple descendants
  - terminal nodes: data points
- Edges represent the relationship



https://www.statisticshowto.datasciencecentral.com/hierarchical-

Dendrogram visualize group memberships

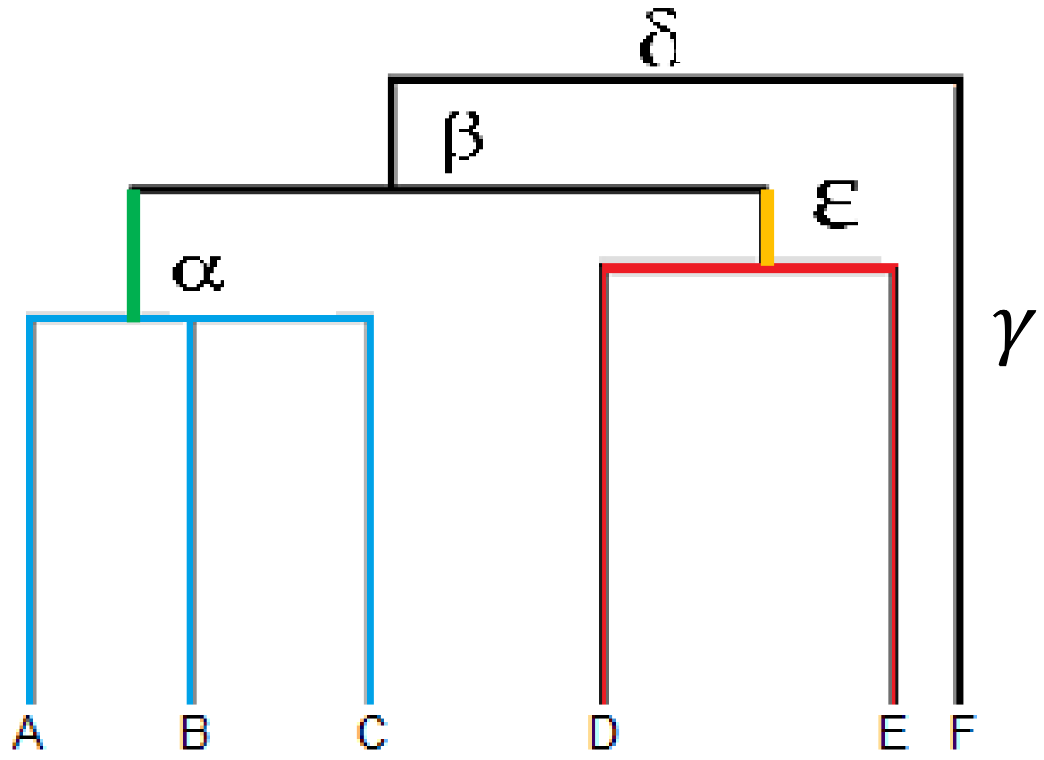
- Leaves A, B, and C are more similar to each other than to D, E, or F.
- Leaves D and E are more similar to each other than to A, B, C, or F.
- Leaf F is different from all of the other leaves.



https://www.statisticshowto.datasciencecentral.com/hierarchical-

Dendrogram visualizes distances among objects

- indicates how A, B, C is different from each other
- indicates how D, E is different from each other
- (α) indicates how the clade of A, B, C is from the rest
- (ϵ) indicates how the clade of D, E is different from the rest.
- \+ ($\alpha + \epsilon$) shows how different the clade of A, B, C is from that of D, E

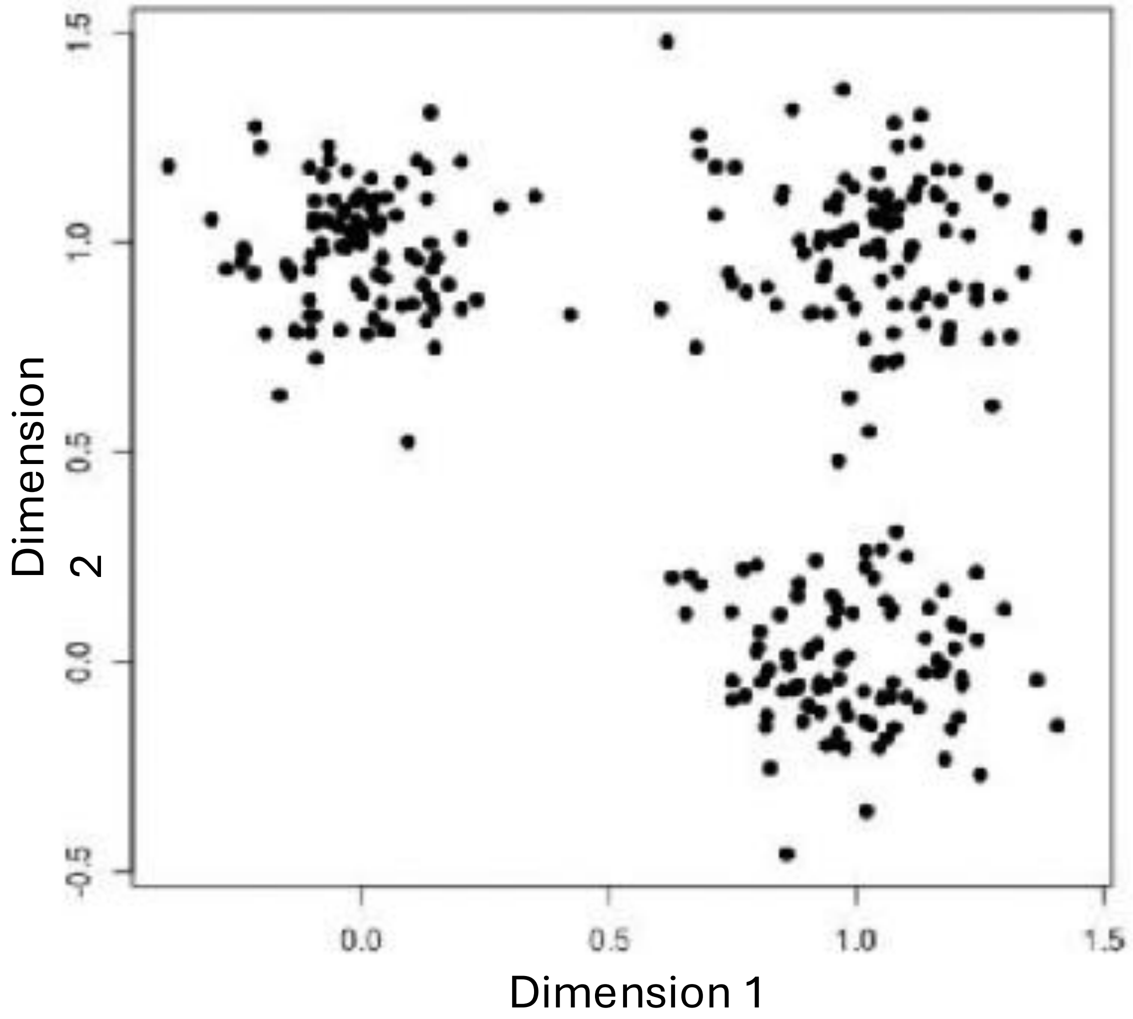


https://www.statisticshowto.datasciencecentral.com/hierarchical-

Clustering based on distance

Algorithm of K-means clustering on 2-D data

- Given data of **two** dimensions (features)

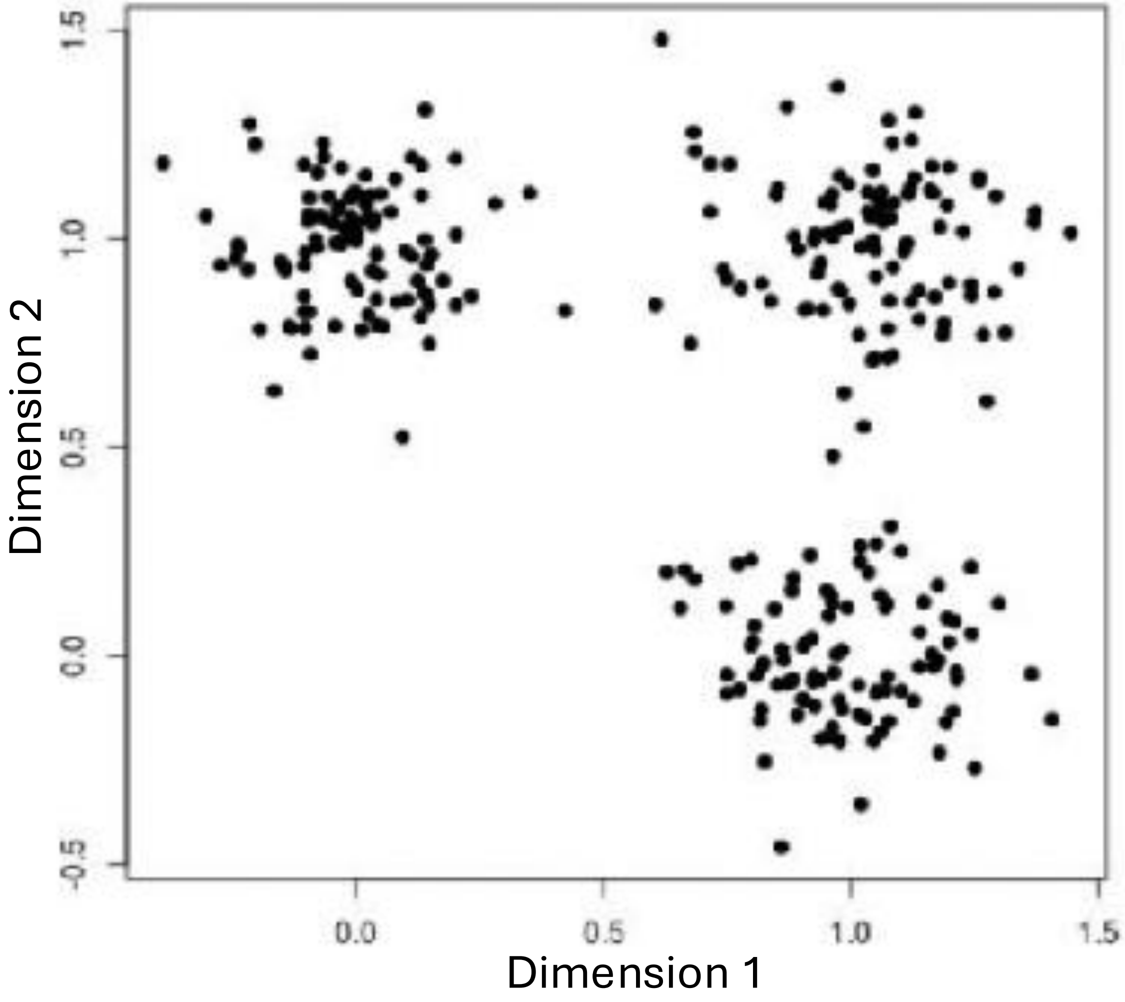


https://towardsdatascience.com/cluster-analysis-create-visualize-and-interpret-customer-segments-

Clustering based on distance

Algorithm of K-mean clustering on 2-D data

- Given data of **two** dimensions
- Assuming **three** clusters (K=3)

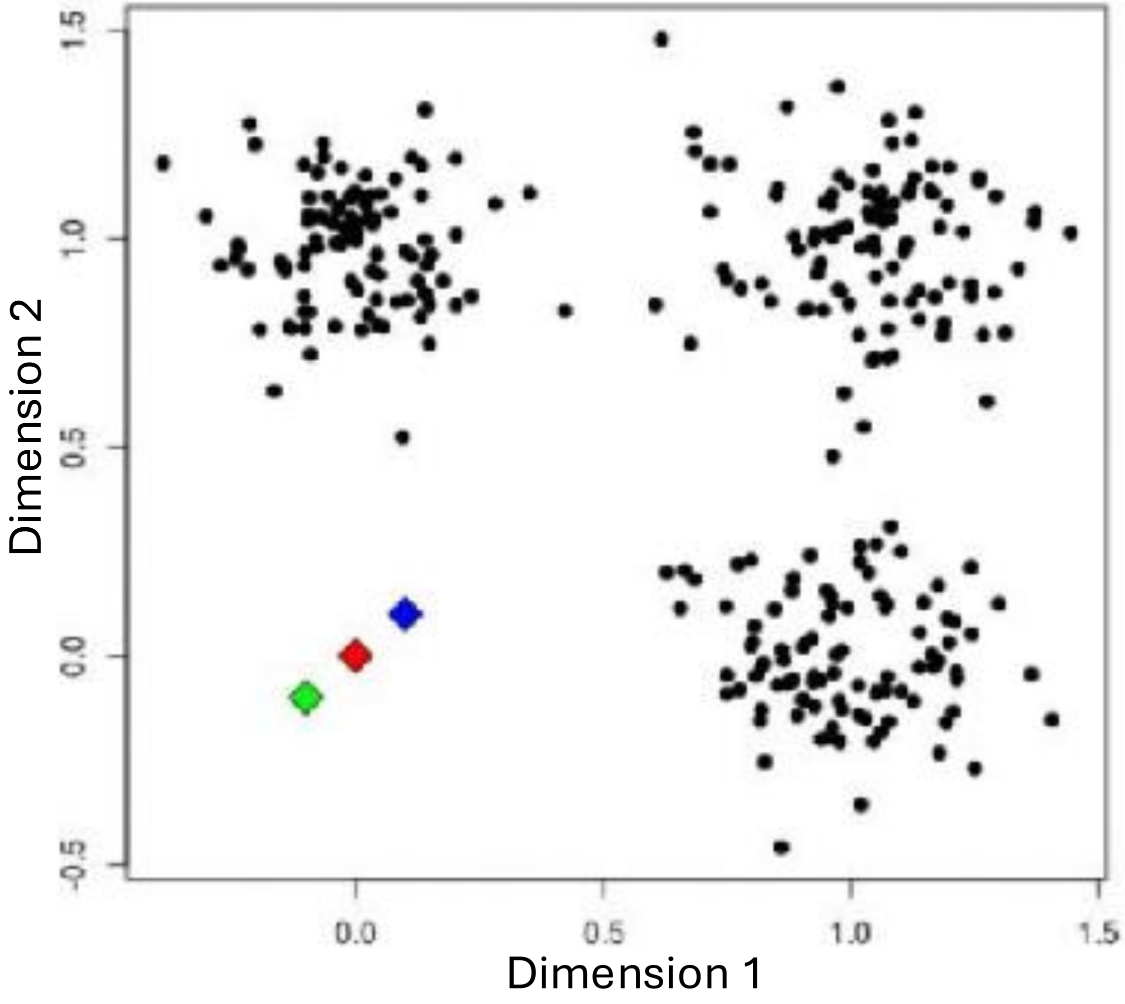


https://towardsdatascience.com/cluster-analysis-create-visualize-and-interpret-customer-segments-

Clustering based on distance

Algorithm of K-mean clustering on 2-D data

- Given data of **two** dimensions
- Assuming **three** clusters (K=3)
- **Randomly** choose the center point (centroid) for the clusters

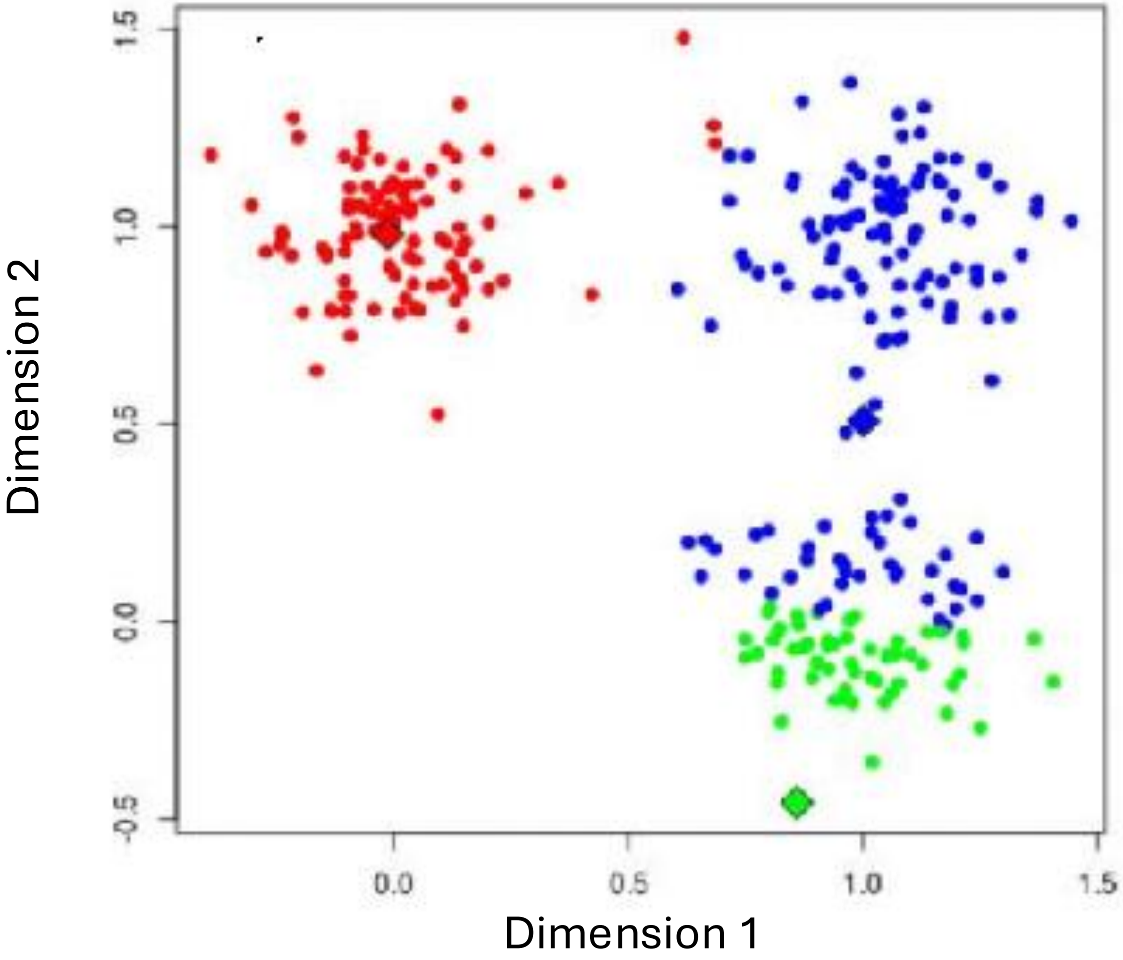


https://towardsdatascience.com/cluster-analysis-create-visualize-and-interpret-customer-segments-

Clustering based on distance

Algorithm of K-mean clustering on 2-D data

- Given data of **two** dimensions
- Assuming **three** clusters (K=3)
- Update cluster assignments based on the **distance**



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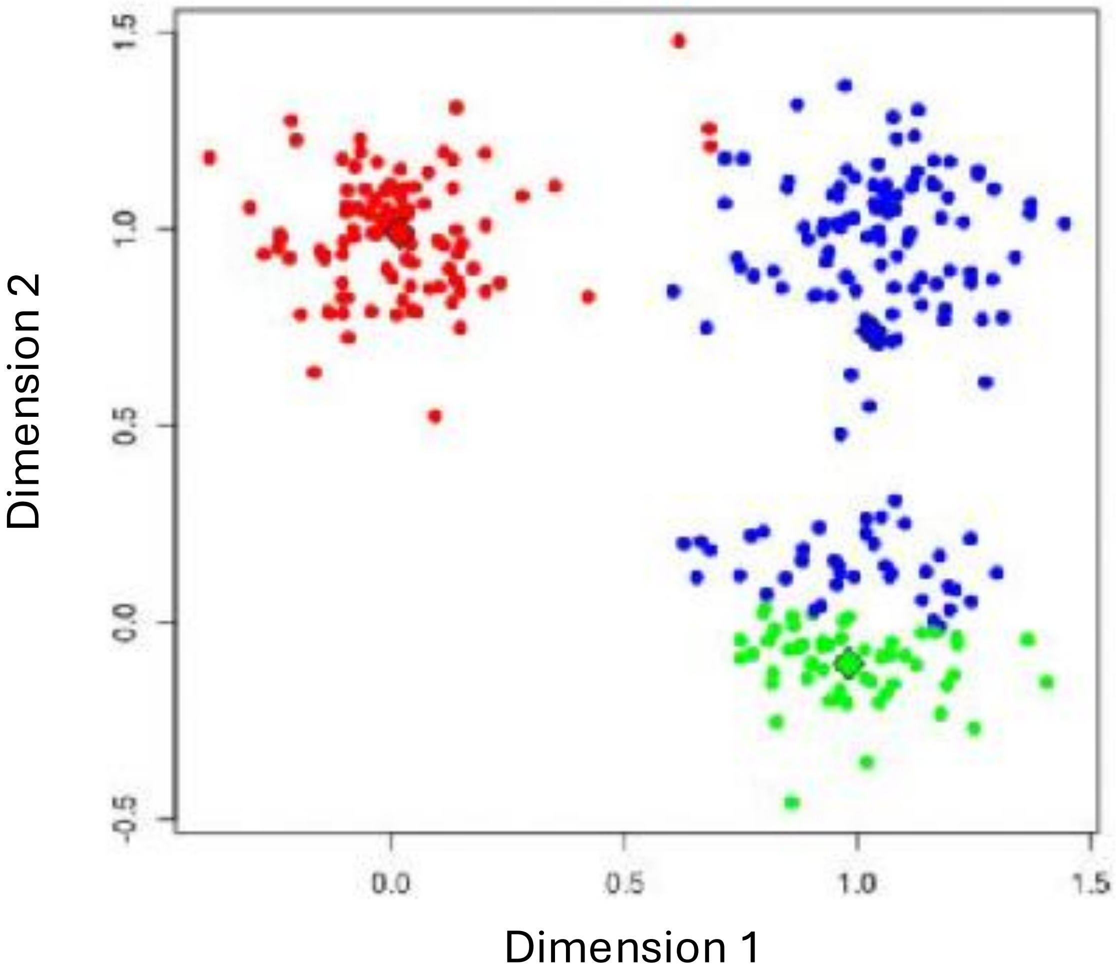
Clustering based on distance

Algorithm of K-mean clustering on 2-D data

- Given data of **two** dimensions
- Assuming **three** clusters (K=3)
- Update the cluster centers



https://towardsdatascience.com/cluster-analysis-create-visualize-and-interpret-customer-segments-

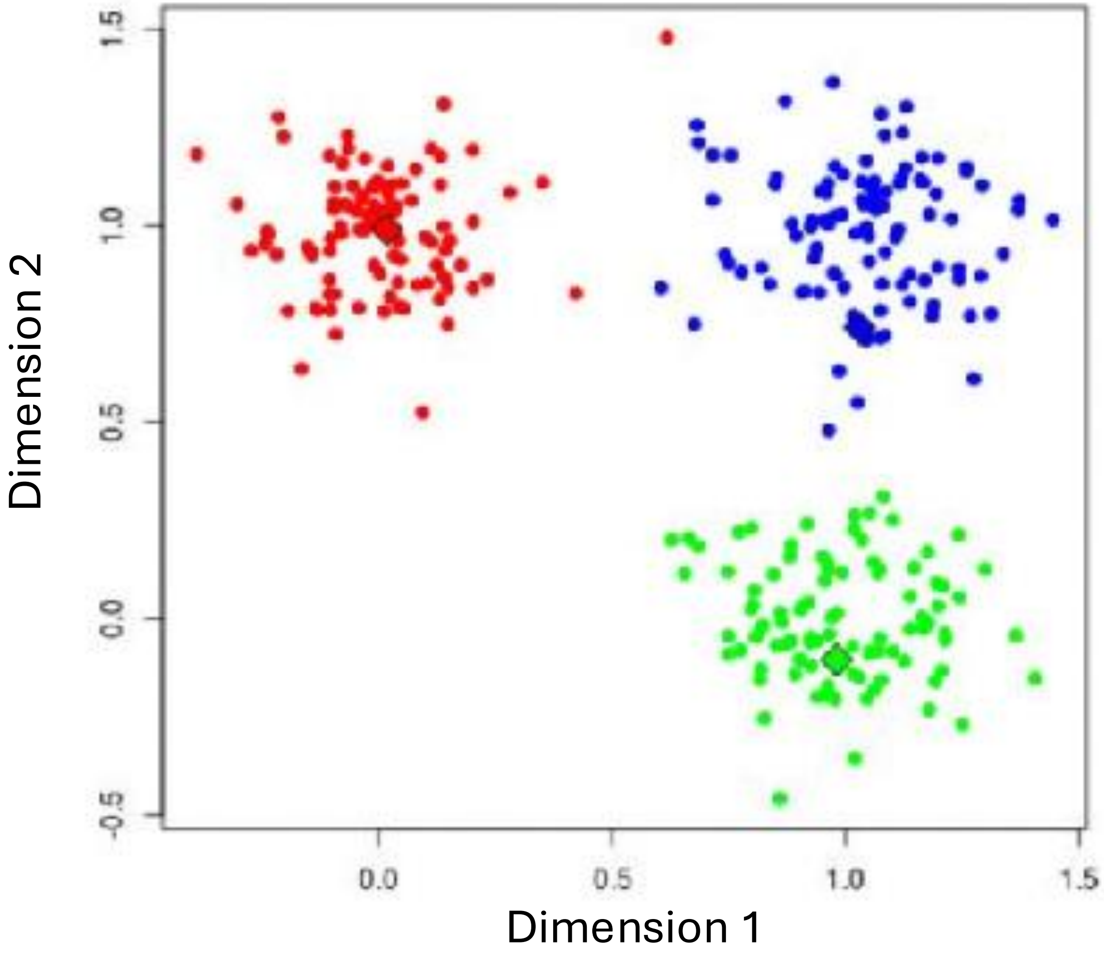
Clustering based on distance

Algorithm of K-means clustering on 2-D data

- Given data of **two** dimensions
- Assuming **three** clusters (K=3)
- Update cluster assignments based on the **distance**

Things to control

- How to determine K
- Location of the initial points
- When to stop
- **How to calculate distance**



https://towardsdatascience.com/cluster-analysis-create-visualize-and-interpret-customer-segments-

Dimension reduction and clustering example

See exercises 9-10

How “unsupervised”?

The models don’t decide everything for you!

- Pre-process the data?
- Feature selection?
- Linear or non-linear?
- What distance metric? (What weights?)
- How many clusters?
- Etc.

Supervised ML

Type of ML	Kind of question	You are given	Goal	Assumptions
Supervised	Is there a mapping from X→Y?	• Input variables (X) • Target/outcome/labels (Y)	• Predict Y • Estimate relationships between X and Y	• The predictive structure (model type) • Signal in X could explain Y
Unsupervised	What structure exists in X?	• Only the data (X) • No outcome/labels	• Describe variation • Find groups, dimensions, gradients, etc.	• Continuous or discrete structure • Local or relational structure

What can you predict with ML?

There are many ways to conceptually organize ML methods.

Most either

- predict a continuous outcome (like regression) or
- predict a discrete outcome (classification)

We could also organize by "function"

- Regression/linear models
- Kernel methods
- Instance-based and distance-based methods
- Tree-based methods
- Neural nets
- Probabilistic/generative models

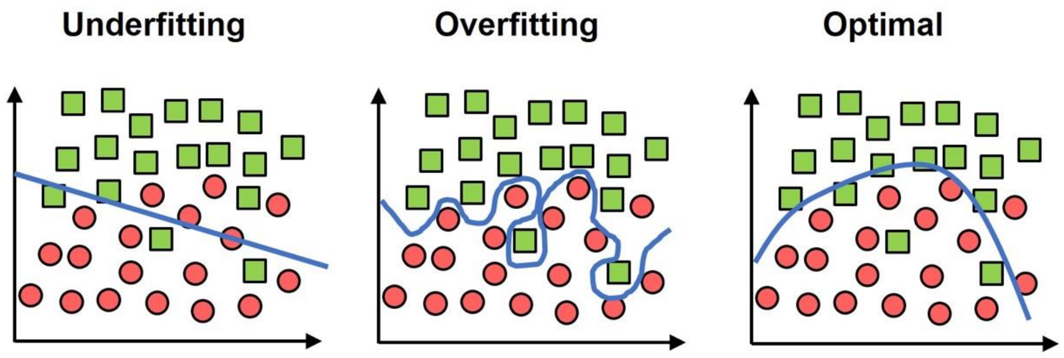
(Note that this is not exhaustive or exclusive.)

Regression/linear models

- Generalized linear (mixed) models
  - Includes regression (linear, logistic)
  - Always linear with respect to predictors
  - You are familiar with many of these
  - Problem: what if n << p?
- → Regularized linear models
  - Ridge regression
  - Lasso regression
  - Elastic net

Problems

- Problem: overfitting
  - = fitting the training data too closely (including noise)
  - = training error << testing error
  - very easy to overfit when n << p
  - → model doesn't generalize
- One solution (for regression): regularization/feature selection
- Note that over- and under-fitting is a universal problem in ML (not just for regression)



Fitting and "bias-variance tradeoff"

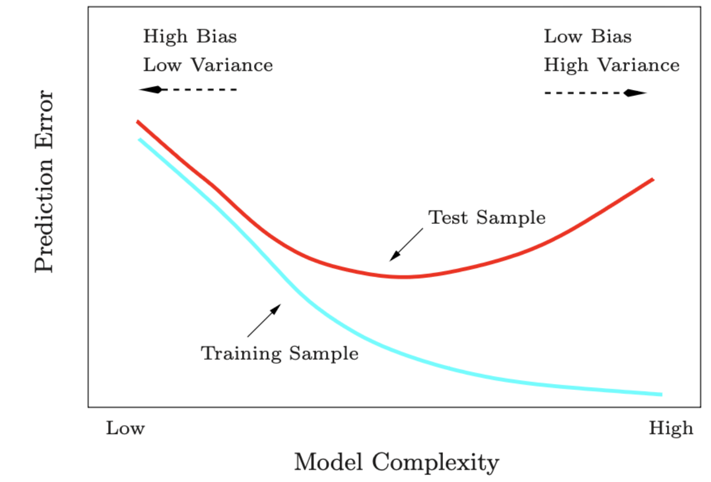
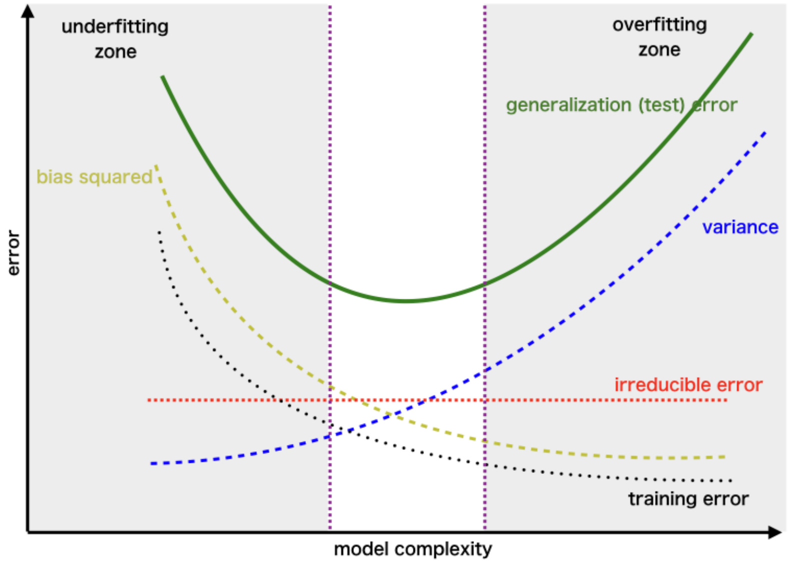


FIGURE 2.11. *Test and training error as a function of model complexity.*



$$\mathrm{MSE} \triangleq \mathbb{E}\left[\left(y - \hat{f}(x)\right)^2\right] = \mathrm{Bias}\left[\hat{f}(x)\right]^2 + \mathrm{Var}\left[\hat{f}(x)\right] + \sigma^2$$

- We want a model that generalizes to new data
- We can decompose test error into bias^2 + variance + irreducible noise

Bias
- Systematic error
- How far is the average prediction from the truth?
- High bias → model too simple (underfitting)

Variance
- Sensitivity of the model to the training data
- How much would prediction change if you trained on a different sample?
- High variance → model too complex (overfitting)

Goal: find the "sweet spot"

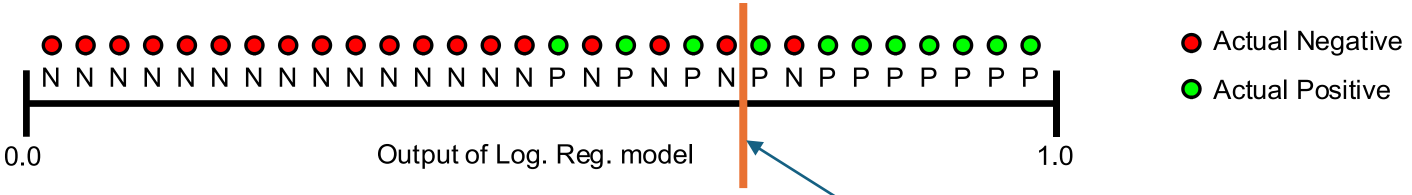
Comparison of regression methods

Method	Model form	What is optimized?	Description	Assumptions
Linear regression	$Y=X\beta+\varepsilon$	$RSS(\beta) = \lVert Y-X\beta \rVert^2$	Simple, familiar	$\varepsilon \sim N(0,\sigma^2)$
GLMMs • Includes linear regression • Includes logistic regression	$g(E[Y\|b])=X\beta+Zb$	Marginal likelihood	Simple, adaptable	• $b \sim N(0,\Sigma)$ • link function g • Conditional independence on random effects • Exponential family distribution for Y
Ridge regression	$Y=X\beta+\varepsilon$	$RSS(\beta) + \lambda \lVert\beta\rVert_2^2$	• Keeps all variables • Shrinks betas • Better estimates for correlated variables	$\varepsilon \sim N(0,\sigma^2)$
LASSO	$Y=X\beta+\varepsilon$	$RSS(\beta) + \lambda \lVert\beta\rVert_1$	• Shrinks betas • Sparse/feature selection: sets many to 0	$\varepsilon \sim N(0,\sigma^2)$
Elastic net	$Y=X\beta+\varepsilon$	$RSS(\beta) + \lambda_1 \lVert\beta\rVert_1 + \lambda_2 \lVert\beta\rVert_2^2$	Blend of ridge and LASSO	$\varepsilon \sim N(0,\sigma^2)$

Regression example

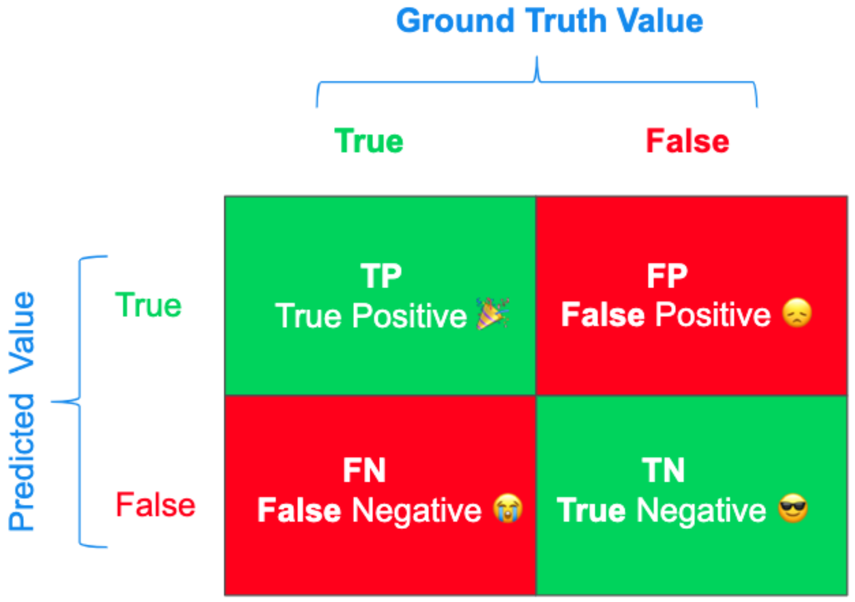
- See Example 1
- What’s wrong here??

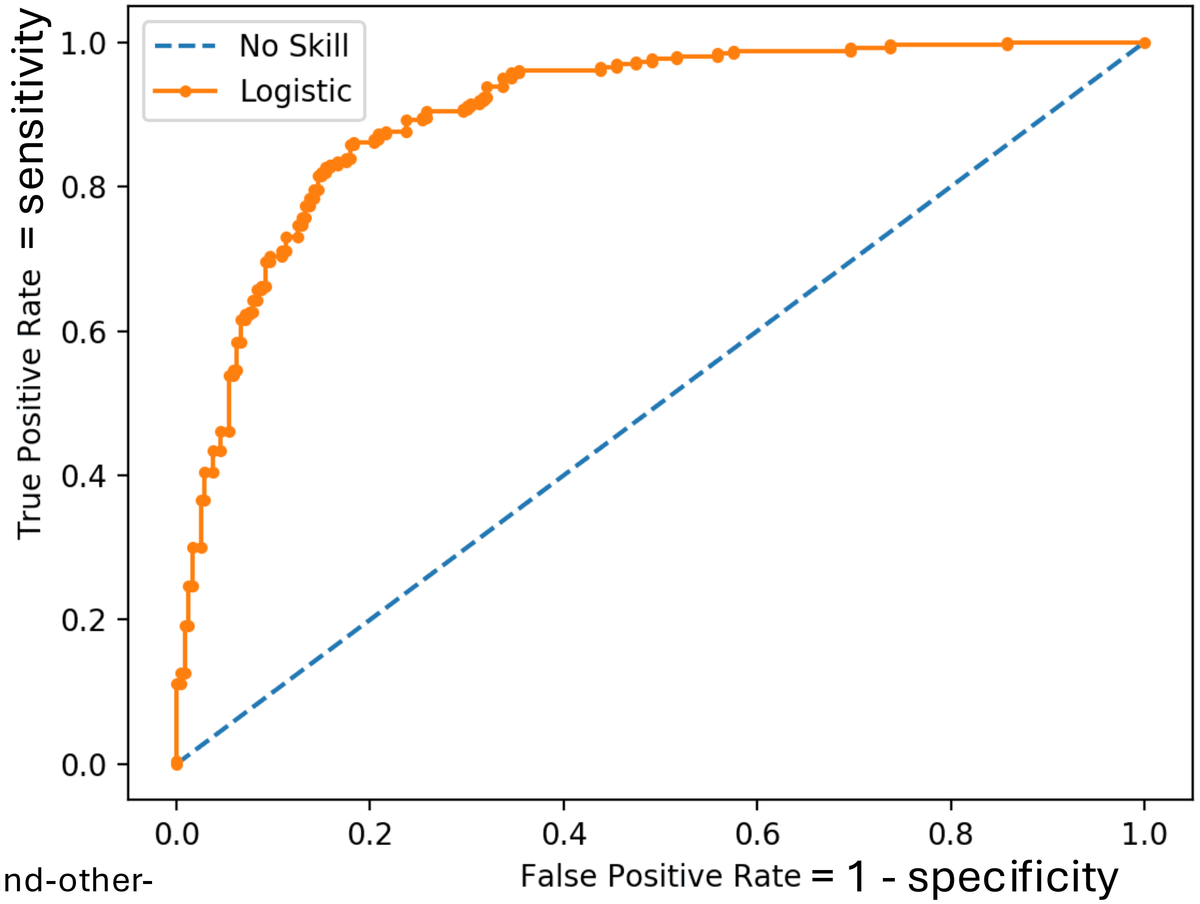
Digression on visualizing classifier results



One possible threshold

- Receiver operating characteristic (ROC)
- The relationship between false positive rate and true positive rate as in a model





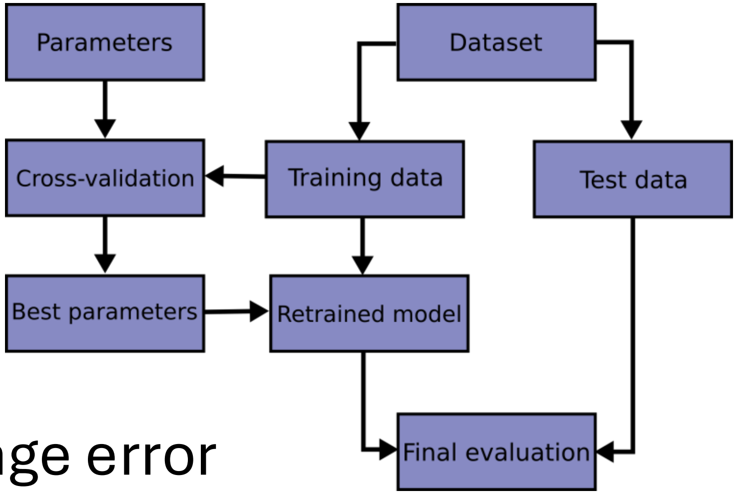
https://blog.floydhub.com/a-pirates-guide-to-accuracy-precision-recall-and-other-

Problems

- Problems
  - We only saw how well the model can do on the training data
  - Training performance overestimates testing performance
  - So how can we estimate the generalization error?
  - How do we choose hyperparameters (like lambda)?
- Solution: cross-validation
- “Bias-variance tradeoff”

Cross-validation

- Split data into training and testing sets (80% vs. 20%, often)
- Make a “grid” of possible values for the hyperparameter(s) (e.g., lambda)
- For each possible value of the hyperparameter(s)
  - Split training data into k folds (often 5 or 10)
  - For each of the k folds
    - Train model on the other k-1 folds
    - Compute its error on the left-out fold
  - Now average the k errors
- Select the value of the hyperparameter with the smallest average error
- Use that value and train the model on the full training set
- Calculate its error on the testing set

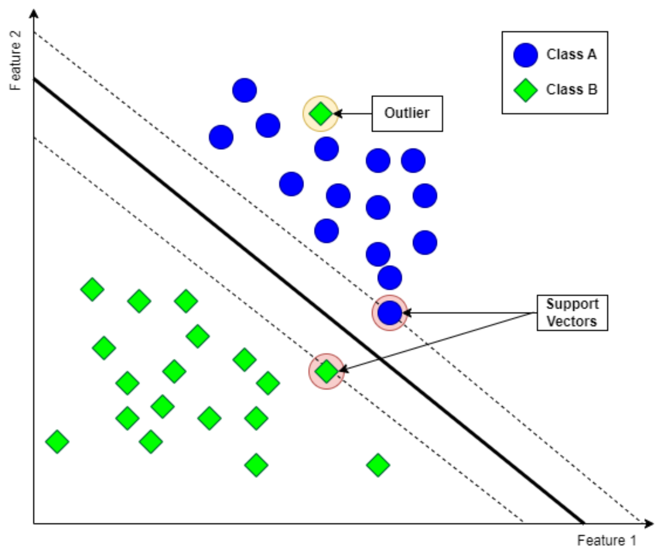


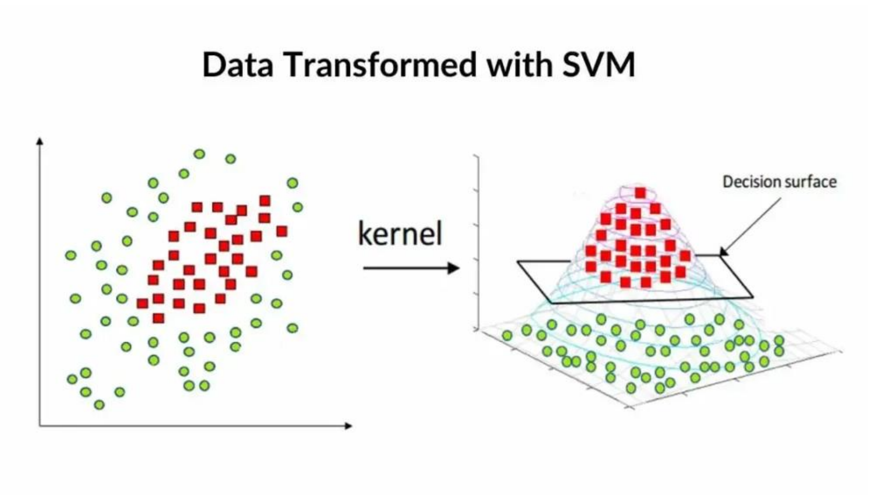
Cross-validation

- See example 2

Kernel methods

- Example: support vector machine
- Goal: binary classification (like logistic regression)
- Want a decision boundary dividing the two classes
- SVM: choose the separating hyperplane with the largest margin





Instance/distance-based methods

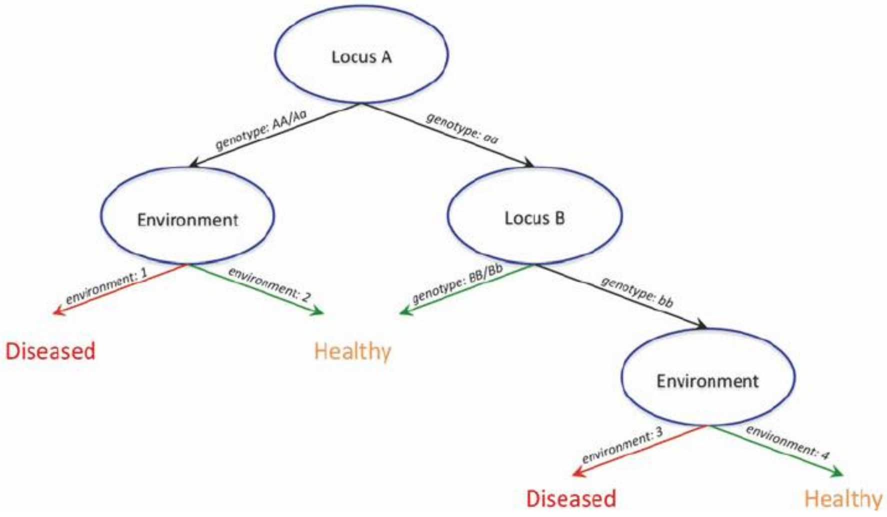
- Distance-based methods: prediction relies on distance/similarity
- Instance-based methods: are training data stored in the model and use for prediction?

k-nearest neighbors (kNN) is an example of both

kNN

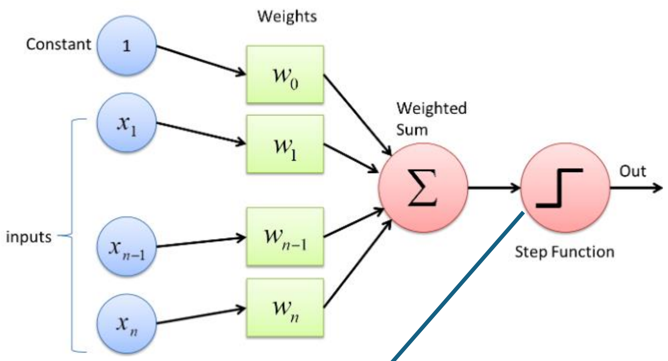
- Idea: classify a new point based on labels of its neighbors
  - Choose k
  - Compute its distance from all training data points
  - Find the k nearest neighbors
  - Assign the point to the majority class among those neigbors
- Parameters to tune
  - k (use cross-validation!)
    - large k = underfitting (low bias and high variance)
    - small k = overfitting (higher bias and lower variance)
  - distance metric
  - feature scaling
  - weighting
  - (dimensionality reduction?)
- See example 3

Tree-based methods

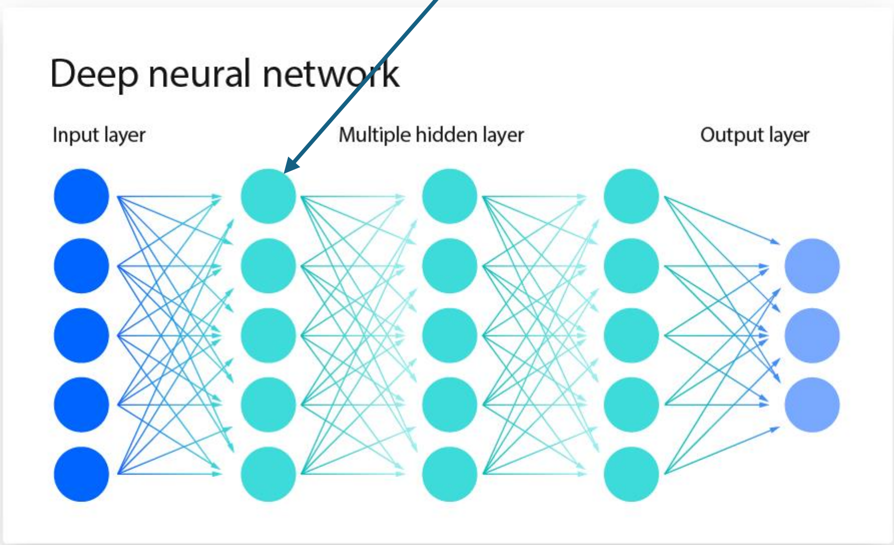


- Decision trees
  - Recursively partition feature-space (many if-then splits)
  - Prediction can be regression-based or majority-vote
  - Advantages
    - Nonlinear
    - Interactions
    - Mixed data types
    - Feature selection
  - Disadvantages: high variance/unstable/over-fitting
  - Parameters: maximum depth, number of samples in each split, number of terminal nodes, penalty for number of nodes
- Random forest
  - Resample (boostrap) many times from your data
  - Train a decision tree on each one and keep all of them
  - Classifier: average the votes of all the trees
- Many extensions/variations exist

Neural nets



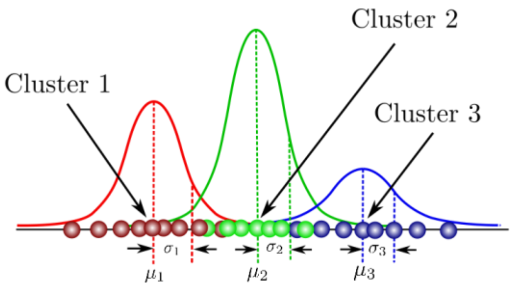
- Compositions of linear and nonlinear transformations
  - Linear transformations of signals between layers
  - Nonlinear activations
- Some extensions
  - Convolutional (CNNs)
    - Often applied to image data
  - Recurrent (RNNs)
    - Often applied to time series data
  - Transformers
    - Include LLMs
  - Autoencoders
    - For unsupervised learning



Probabilistic models

Core idea

- "How likely are the data to have been generated by each class?"
- Prediction: pick the class that makes the data most probable



Examples

- Naïve Bayes
  - " How strongly does each piece of evidence (e.g., dosage of allele A, expression of gene B, clinical feature C) point to "case vs. control"?
  - Naïve = no interactions
- Gaussian mixture classifiers

- Hidden Markov models
  - For sequential data (e.g., chromatin states along a segment of a chromosome)
  - You observe noisy signals but cannot directly observe state
  - Infer the state most likely to have emitted a signal